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LEICESTER

Simulation Tool for Economic Crisis Prediction

Using Inflation and Macroeconomic Indicators

**A dissertation report submitted to the University of Leicester for my
master's degree course in 'AI for Business Intelligence'**

By

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ABSTRACT

In this research, a data-driven pipeline for inflation forecasting and economic crisis risk recognition in the United Kingdom with key macroeconomic indicators is implemented. The last few years have revealed the significant fluctuations in inflation rates, especially in the wake of the COVID-19 pandemic and the unprecedented surge in prices experienced in 2022-2023. Such structural breaks and non-linear economic characteristics often make it difficult to use traditional linear time-series models, motivating the need to create more powerful predictive frameworks as an essential policy requirement.

In response to this, we have now created a dataset of key macroeconomic data for the last 25 years of the data series, combining the Gross Domestic Product (GDP), unemployment rates, retail sales index (RSI), Bank of England base rate, and indicators of the FTSE100. The stationarity properties and monetary policy relevance of Year-on-Year (YoY) Inflation make it our target variable. Feature engineering is customised to focus specifically on the persistence of inflation in the past and other macroeconomic relationships.

We use and compare three different modelling techniques: ARIMA is used as a basis of comparison for a standard statistical technique, Random Forest is the main machine learning technique used, and XGBoost is a top-tier gradient boosted ensemble model. To fully test temporal robustness, model performance is evaluated in several validation windows by Root Mean Square Error (RMSE) and the coefficient of determination (R^2). The Random Forest model has the lowest RMSE value, around 2.64, and the highest accuracy, which shows its capability of mapping the non-linear economic interactions, empirically.

Multi-period validation, however, indicates that all three architectures underestimate the actual inflation rates in periods of extreme and anomalous inflation; underpredicting during the regime shift of 2022–2023. This weakness implies the limitations of solely data-driven models at structural economic breaks. This study proposes a framework for detection of crisis events based on a threshold to make up the deficit between technical forecasting and operational decision making by converting the numerical outputs of inflation into risk categories (Low, Moderate, High, Crisis). This framework is embedded in a tool that is an interactive dashboard and simulation of scenarios of future inflation risks in various economic contexts, where the user can make choices and simulate the risks that inflation faces in different economic scenarios.

This study integrates three elements: forecasting, multi-window validation, and interactive simulation, presenting the predictive data from the three elements as a coherent decision-support system, rather than three separate ones converting the raw forecasting data into macroeconomic insights.

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I am deeply grateful to all the lecturers and staff of the University of Leicester for their teaching, guidance, and support throughout my studies in the MSc Artificial Intelligence for Business Intelligence programme. The knowledge and skills gained during the course, particularly in the areas of data analytics, machine learning, forecasting, business intelligence, and data visualisation, provided the foundation for the successful completion of this dissertation.

This research project provided an opportunity to apply analytical and technical concepts to a real-world economic forecasting problem. Through the development of an inflation forecasting framework and an interactive economic crisis simulation tool, I was able to strengthen my understanding of predictive analytics, machine learning, economic modelling, data visualisation, and decision-support systems.

Finally, I would like to thank my classmates, colleagues, and friends for their encouragement, support, and companionship throughout this academic journey. Their motivation and positive influence made this experience both enjoyable and rewarding.

I am sincerely grateful to everyone who contributed, directly or indirectly, to the successful completion of this dissertation.

DECLARATION

I hereby would like to declare that the information in this document is original, and any information gathered from books, websites or any other sources are clearly mentioned in the references with author names.

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Table of Contents

Simulation Tool for Economic Crisis Prediction	
ABSTRACT.....	i
ACKNOWLEDGEMENTS	ii
DECLARATION.....	iii
Table of Contents	iv
Table of Figures.....	vii
List of Tables.....	viii
1. INTRODUCTION.....	1
1.1 Background of Study	1
1.2 Problem Statement.....	1
1.3 Research Aim & Objectives	2
1.4 Research Question	2
1.5 Scope of Study	2
1.6 Structure of the Report	2
2. LITERATURE REVIEW.....	4
2.1 Introduction.....	4
2.2 Time Series Analysis in Economic Forecasting	4
2.3 Traditional Time Series Models for Forecasting.....	4
2.3.1 ARIMA Model.....	4
2.3.2 SARIMA Model.....	5
2.3.3 Limitations of Traditional Models	5
2.4 Machine Learning Approaches in Economic Forecasting.....	5
2.4.1 Advantages of Machine Learning in Economic Modelling	6
2.4.2 Random Forest in Economic Forecasting	6
2.4.3 XGBoost in Economic Forecasting	6
2.4.4 Comparison of Machine Learning and Traditional Models	7
2.4.5 Limitations of Machine Learning in Economic Forecasting	8
2.4.6 Relevance to This Study	8
2.5 Hybrid and Comparative Modelling Approaches.....	8
2.6 Challenges in Inflation Forecasting.....	9

2.7	Research Gap	11
2.8	Contribution to this Study	11
2.9	Chapter Summary	13
3.	METHODOLOGY	14
3.1	Introduction.....	14
3.2	Research Design	14
3.3	Data Collection	15
3.4	Data Preprocessing	17
3.5	Feature Engineering	18
3.6	Exploratory Data Analysis Methodology.....	20
3.7	Forecasting Models	21
3.7.1	ARIMA Model.....	21
3.7.2	Random Forest.....	22
3.7.3	XGBoost	22
3.8	Model Evaluation Metrics.....	23
3.9	Validation Strategy	24
	Holdout Validation.....	25
	Walk-Forward Validation	26
3.10	Crisis Detection Framework.....	26
3.11	Dashboard and Simulation Tool.....	27
	Power BI Dashboard.....	27
	Streamlit Simulation Tool.....	27
	Integration with Forecasting Framework	28
3.12	Chapter Summary	28
4.	EXPLORATORY DATA ANALYSIS.....	30
4.1	Introduction.....	30
4.2	Inflation Trend Analysis	30
4.3	Stationarity Analysis.....	32
4.4	Distribution Analysis	33
4.5	Correlation Analysis	35
4.6	Lag Relationship Analysis	37
4.7	Crisis Period Behaviour Analysis	39
4.8	Key Insights from EDA	40

4.9	Chapter Summary	40
5.	MODEL IMPLEMENTATION, RESULTS AND VALIDATION.....	42
5.1	Introduction.....	42
5.2	ARIMA Implementation and Results.....	43
5.3	Random Forest Implementation and Results.....	45
5.4	XGBoost Implementation and Results.....	48
5.5	Comparative Performance Analysis.....	50
5.6	Validation and Robustness Analysis.....	52
5.7	Structural Break and Crisis Forecasting Discussion	54
5.8	Chapter Summary	55
6.	DASHBOARD AND SIMULATION TOOL	57
6.1	Introduction.....	57
6.2	Power BI Dashboard Implementation	58
6.3	Streamlit Simulation Framework.....	60
6.4	Practical Applications of the System	63
6.5	Dashboard and System Limitations	64
6.6	Chapter Summary	64
7.	CONCLUSION AND FUTURE WORK	66
7.1	Research Summary	66
7.2	Achievement of Research Objectives	66
7.3	Key Findings.....	67
7.4	Limitations of the Study	68
7.5	Future Work	68
7.6	Final Conclusion.....	69
	LINKEDIN POSTS.....	70
	Post 1	71
	Post 2	72
	Post 3	73
	REFERENCES.....	74
	APPENDIX.....	76
	Table A1. Data Sources.....	76
	Table A2. Engineered Features	76
	Table A3. Final Model Parameters.....	76

Table of Figures

Figure 1: Comparison between Traditional Statistical Models and Machine Learning Approaches	7
Figure 2: Comparative Forecasting Framework Used in the Study	9
Figure 3: UK Inflation Dynamics and Crisis Periods (2000–2024)	10
Figure 4: Proposed Forecasting and Decision-Support Framework	12
Figure 5: CPI to Inflation_YoY Transformation	17
Figure 6: Feature Engineering Framework	19
Figure 7: Exploratory Data Analysis Workflow	20
Figure 8: Walk-Forward Validation Process	26
Figure 9: Exploratory Data Analysis Workflow	27
Figure 10: Simulation Workflow	28
Figure 11: UK Inflation Trend (2000–2024).....	31
Figure 12: Consumer Price Index (CPI) Trend (2000–2024)	32
Figure 13: Distribution of Inflation_YoY	34
Figure 14: Distribution of Selected Macroeconomic Variables	35
Figure 15: Correlation Heatmap of Macroeconomic Variables	36
Figure 16: Lag Correlation Heatmap	38
Figure 17: UK Inflation Dynamics During Crisis Periods.....	39
Figure 18: ARIMA Actual vs Predicted Inflation	43
Figure 19: Random Forest Actual vs Predicted Inflation.....	45
Figure 20: Random Forest Feature Importance	47
Figure 21: XGBoost Actual vs Predicted Inflation	49
Figure 22: Comparative Forecasting Performance Across Models	51
Figure 23: Actual vs Predicted Inflation Across Validation Windows.....	53
Figure 24: Power BI Dashboard Architecture.....	58
Figure 25: Power BI dashboard interface	59
Figure 26: Power BI dashboard interface (Crisis Simulator).....	60
Figure 27: Streamlit Simulation Framework Architecture.....	61
Figure 28: Main Interface of Streamlit Inflation Risk Simulation Dashboard	62
Figure 29: Dashboard Scenario/Risk Simulator	62
Figure 30: LinkedIn Post 1	71
Figure 31: LinkedIn Post 1	71
Figure 32: LinkedIn Post 2	72
Figure 33: LinkedIn Post 2	72
Figure 34: LinkedIn Post 3	73
Figure 35: LinkedIn Post 3	73
Figure 36: A1. Final Streamlit-based Economic Crisis Prediction Dashboard.....	77

List of Tables

Table 1: Key Differences between Traditional Time Series Models and Machine Learning Techniques for Inflation Forecasting	7
Table 2: Summary for Macroeconomic variables	16
Table 3: Engineered Features and Purpose	19
Table 4: Risk Classification based on Inflation Level	27
Table 5: ARIMA Forecasting Performance Summary	44
Table 6: Random Forest Forecasting Performance Summary	46
Table 7: XGBoost Forecasting Performance Summary	49
Table 8: Comparative Forecasting Performance of Models	50
Table 9: Validation Window Configuration	53
Table 10: Inflation Risk Classification Framework	59
Table 11: A1. Data Sources	76
Table 12: A2. Engineered Features	76
Table 13: A3. Final Model Parameters (RF)	76
Table 14: A3. Final Model Parameters (XGB)	76

1. INTRODUCTION

1.1 Background of Study

Inflation is one of the primary indicators of macroeconomic health and has an immediate indirect effect on monetary policies of the central bank, the price of financial products, and people's real purchasing power. The central banks, such as the Bank of England, usually prescribe a target inflation rate to ensure sustainable economic growth, typically 2%. But the world has become a much more turbulent place. The longstanding price stability is severely disrupted by fractured supply chains, geopolitical realignment and macroeconomic adjustments after the pandemic.

The current inflationary period from 2020 to 2023 is a fundamental structural change in the inflation process. Early stages of the COVID-19 pandemic muted demand, and then once more in late 2022, as supply was clogged, inflation reached unprecedented levels. The abrupt changes in regime significantly undermine traditional econometric models, which rely on the assumption that linear models from the past will still hold true for the future.

The time-series forecasting has been a time-honoured tool that has been used by economists to manoeuvre these patterns. Classical models, such as ARIMA, are very successful at dealing with stable environments, but are unable to tackle the non-linear, recursive dynamics of today's market economies. For this reason, machine learning has become an important tool with its flexibilities in pattern recognition without any specific structure.

At the same time, contemporary economic control requires moving from a numerical forecasting system to a decision support system with dynamic architectures. In parallel, contemporary economic management requires the shift from static numeric forecasting to dynamic decision support architectures. But policymakers and financial analysts need more than just future values—they need early warning indicators as well as risk classifications to help them strategy ahead of the curve.

1.2 Problem Statement

While significant progress has been made in the field of architectural developments of predictive modelling, making accurate predictions of inflation during periods of high systemic instability is still a challenging task. Traditional statistical tools fail to perform well during structural change, and flexible machine learning tools often struggle to transfer cleanly when moving from one completely different economic regime to another.

While there is much work out there on prediction accuracy, there is also a significant gap in the literature in terms of how algorithmic results become usable, real-world

decision-making tools. Most of the research emphasizes optimization metrics, with little attention to the downstream application. This is reflected in three clear, systemic limitations: In particular:

- It is well known that extreme non-linear inflation increases are underestimated or lag in all predictive models.
- Even with complex algorithmic outputs, it is not possible to work safely in policymaking without technical people.
- Forecasting pipelines do not have formal, automated linkages with systemic crisis detection and early-warning systems.

1.3 Research Aim & Objectives

Aim:

To develop a robust inflation forecasting framework that integrates macroeconomic indicators and converts predictions into actionable economic crisis signals.

Objectives:

- i. To analyse the relationship between the macroeconomic indicators and inflation in the UK economy.
- ii. To build forecasting models using statistical and machine learning techniques.
- iii. To allow comparison of model performance across evaluation periods.
- iv. To test the robustness of models in stable and volatile economic regimes.
- v. To develop a framework for crisis detection based on inflation thresholds.
- vi. Develop an interactive dashboard and simulation tool for decision support.

1.4 Research Question

To what extent do macroeconomic indicators predict inflation and signal economic crisis risk?

1.5 Scope of Study

This study uses monthly macroeconomic data for the UK economy over the period 2000-2024. The analysis covers key indicators like GDP, unemployment, retail sales index (RSI), bank rate and FTSE100. The paper is concerned with the forecasting performance and the ability to detect crises and not with causal inference.

1.6 Structure of the Report

This dissertation is structured as follows:

- Chapter 2: Literature Review
- Chapter 3: Methodology
- Chapter 4: Exploratory Data Analysis
- Chapter 5: Model Implementation, Results and Validation

- Chapter 6: Dashboard and Simulation Tool
- Chapter 7: Conclusion and Future Work

2. LITERATURE REVIEW

2.1 Introduction

Inflation forecasting plays a key role in macroeconomic policy making, stability of financial markets and economic decision making. Good inflation predictions are essential for central banks like the Bank of England when they determine the monetary policy instruments that they use, such as interest rates and quantitative easing. However, recent structural changes in the global economy, including the COVID-19 pandemic, supply chain disruptions and geopolitical tensions, have made inflation forecasting considerably more challenging.

The assumptions of traditional economic models are that macro-economic variables such as GDP growth, unemployment and inflation are relatively stable. However recent empirical studies indicate that these relationships may in fact be non-linear and change in regime. These have also led to the growing interest in data-driven and machine-learning methodologies that can capture complex interactions and evolving patterns in economic data.

This chapter reviews the literature related to inflation forecasting under three main headings:

- Traditional time-series models
- Machine learning approaches in economic forecasting
- The role of macroeconomic indicators in predicting inflation

The review also points out the gap between forecasting models and operational decision-making systems. This study aims to fill this gap.

2.2 Time Series Analysis in Economic Forecasting

One of the more popular methods in economic forecasting is time series analysis. A time series consists of data collected in a sequential manner over time, and the correlation over time is crucial in understanding the patterns and trends.

Stationarity is one of the basic assumptions in classical time series modelling and it means that the statistical properties of the series (mean and variance) are constant over time. The macro-economic information, however, is typically non-stationary (e.g., inflation) and requires pre-processing (e.g., differencing or logarithmic), before it can be used for modelling.

2.3 Traditional Time Series Models for Forecasting

2.3.1 ARIMA Model

The Autoregressive Integrated Moving Average (ARIMA) model is one of the most widely used statistical approaches for time series forecasting. It combines three components:

- Auto Regression (AR): dependence on past values
- Integration (I): differencing to achieve stationarity
- Moving Average (MA): dependence on past errors

ARIMA models perform best when the data is stationary and the relationship between variables is linear. Previous studies such as Box and Jenkins (1976) demonstrated the effectiveness of ARIMA models for time-series forecasting under stable conditions. However, later research by Stock and Watson (2008) highlighted that forecasting performance can deteriorate when structural breaks and changing economic relationships occur.

As pointed out in previous studies, ARIMA models are limited in their ability to capture sudden structural breaks, for instance, due to financial crises or pandemic shocks.

2.3.2 SARIMA Model

Seasonal ARIMA, or SARIMA, is an extension of the ARIMA model that takes seasonality in time series data into consideration. Economic forecasting often uses SARIMA models when there is cyclic behaviour in the data. However, SARIMA models can account for seasonality but still are based on linear assumptions and may have problems during times of structural economic disruption and volatility.

2.3.3 Limitations of Traditional Models

Despite their widespread use, traditional time series models suffer from several limitations:

- Assumption of linearity
- Underperform during structural breaks
- Limited ability to model complex interactions between variables
- Dependence on manual parameter tuning

These limitations have encouraged researchers to explore more flexible and data-driven approaches, particularly machine learning models.

2.4 Machine Learning Approaches in Economic Forecasting

The forecasting of economic indicators with the help of machine learning (ML) algorithms has been gaining popularity in recent years. Note that machine learning methods are based on past observations, rather than making assumptions about how the data is supposed to be distributed, as with traditional statistical methods.

Today, the development of computational resources and the expansion of macroeconomic data available for research have enabled the application of ML models to forecast, recession analysis and assessing the financial markets.

One of the typical challenges when applying traditional econometric tools in predicting macroeconomic variables is the inability to capture nonlinearity and

structural breakages as well as high dimensionality. Indeed, in economic systems, a lot of factors play non-linear and complex non-linear relationships, and a lot of factors are policy sensitive, consumer behaviour sensitive and even globally economic environment sensitive.

According to Mullainathan and Spiess (2017), machine learning techniques offer advantages in identifying complex patterns within large datasets that may not be captured by traditional econometric models.

The traditional econometric modelling methods like ARIMA models are not suitable for modelling the nonlinear dependency, however the machine learning approach can easily overcome this problem by finding unobserved patterns in the data.

2.4.1 Advantages of Machine Learning in Economic Modelling

There are several benefits of machine learning in comparison to classical statistics when applied for economic forecasting:

1. Non-linear relationships recognition ability
2. Handling High-Dimensional Data
3. Less assumptions made in models
4. Noise reduction

In addition, ensemble learning techniques such as Random Forest help improve forecasting stability and reduce the impact of noise within economic data.

2.4.2 Random Forest in Economic Forecasting

Random Forest is a widely used ensemble learning technique. In the training phase, it creates several decision trees and combines their predictions for the final prediction. This algorithm takes a random sample of data instances and features, reduces the risk of over-fitting and enhances the stability of the forecasting.

Breiman (2001) introduced Random Forest as an ensemble learning technique capable of improving predictive accuracy while reducing overfitting through the aggregation of multiple decision trees.

The other advantage of Random Forest, over other models, is the capability of modelling more complex interactions between several macroeconomic variables. Because of the many variables that impact inflation including GDP growth, unemployment and interest rates, the model is well suited to capture non-linear economic relationships and changing macro-economic patterns.

2.4.3 XGBoost in Economic Forecasting

XGBoost is a powerful gradient boosting machine learning methodology that incrementally constructs models to counteract the errors of previous models. This kind of iterative learning makes it possible to forecast higher accuracy for complicated data sets while using XGBoost.

Chen and Guestrin (2016) demonstrated that XGBoost can achieve high predictive performance through gradient boosting optimisation while maintaining computational efficiency.

XGBoost is particularly well suited for economic forecasting tasks where it is able to learn more nuanced non-linear relationships and can deal with missing data effectively and for adjusting model performance with regularisation methods. The model's attributes allow for the analysis of complex macro-economic relationships and for the analysis of changing economic circumstances.

2.4.4 Comparison of Machine Learning and Traditional Models

The key differences between machine learning models and traditional time series approaches can be summarised as follows:

Feature	Traditional Models	Machine Learning Models
Assumptions	Strong (stationarity, linearity)	Minimal assumptions
Relationships	Linear	Non-linear
Flexibility	Limited	High
Interpretability	High	Moderate
Performance in volatility	Weak	Stronger but not perfect

Table 1: Key Differences between Traditional Time Series Models and Machine Learning Techniques for Inflation Forecasting

Traditional Programs: Define algo/logic to compute output



Machine Learning: Learn model/logic from data

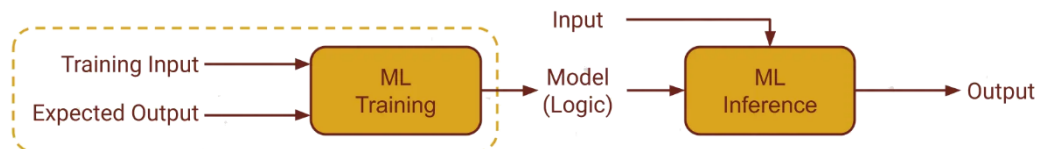


Figure 1: Comparison between Traditional Statistical Models and Machine Learning Approaches

Figure 1 demonstrates the basic distinctions between conventional statistical methods and machine learning techniques. Conventional statistical methods use predetermined logic or structure to derive outcomes based on the inputs provided. On the other hand, machine learning uses a training process to extract patterns from input data and then apply the developed model to derive predictions. Such an approach helps machine learning methods account for the nonlinear associations that may exist between variables. The distinction is particularly useful in economic forecasting where macroeconomic associations are ever-changing and hard to represent using set rules.

Machine learning models usually outperform traditional statistical approaches in predictive accuracy, but often at the expense of interpretability. However, models like

ARIMA are more suitable to explain the forecasting behaviour, but may not be able to capture the complex non-linear relationships of the modern economic system.

2.4.5 Limitations of Machine Learning in Economic Forecasting

Despite of having advantages, ML models come with a few limitations as well:

1. Risk of Overfitting
2. Lack of Interpretability
3. Sensitive to Changes in the Data
4. Underperform in Extreme Events

2.4.6 Relevance to This Study

Considering the constraints associated with conventional models along with benefits offered by machine learning algorithms, this paper employs comparative modelling in its study. In this regard, the ARIMA model is utilized as the benchmark, whereas Random Forest and XGBoost have been implemented for their non-linearity capabilities.

In addition to forecasting, this paper takes the research to another level where it employs a validation scheme for different time periods and uses the model outputs for developing a crisis detection mechanism.

2.5 Hybrid and Comparative Modelling Approaches

In the latest efforts in economic modelling we find that there is no specific modelling approach that can be always best in relation to others, regardless of economic conditions. The classical econometric modelling approach may provide results that are easily interpreted and theoretically supported. Machine learning models, by contrast, accommodate non-linear dynamics without requiring strict a priori structural assumptions. Therefore, most of the research have been focusing on experiments by comparative modelling methods.

Comparative modelling techniques are gaining in popularity in the field of macroeconomic forecasting. That is because the economy is very sensitive to sudden changes in economic structures, policies and behavioural changes. In other words, a model that can work efficiently under normal economic circumstances may not be effective during economic crisis and abnormal volatility.

This has led to the growing use of comparative methods that include comparing a few models with various data and validation periods.

Despite these developments, however, there are still a number of challenges. First, many researchers test and evaluate their models with a single train-test split, which may not capture economic environment changes. Secondly, few works have concentrated on the practice of forecasting results, which entails converting the forecasts to useful indicators of risk.

This research is based on the comparative modelling approach, and is done using:

- ARIMA as a traditional benchmarking model
- Random Forest as the core machine learning model
- XGBoost as an enhanced boosting-based ensemble learning model

More than one model shall be adopted so that a wider range of economic conditions can be evaluated for the accuracy of prediction. Further, this research is not only based on predicting methods but has also included the validation within several periods of analysis and converted the model prediction into crisis risk categories.

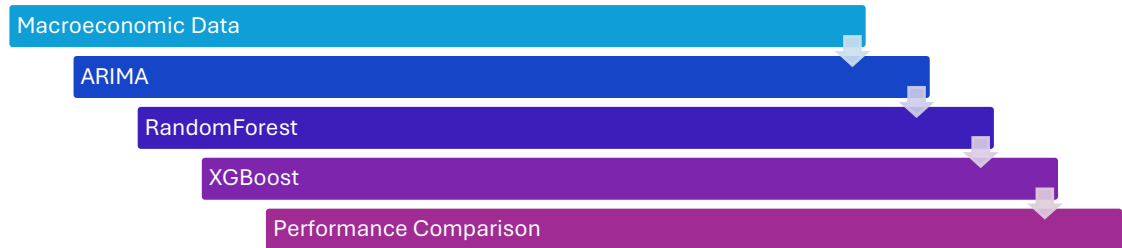


Figure 2: Comparative Forecasting Framework Used in the Study

Figure 2 illustrates the comparative modelling approach used in this study is illustrated. The same macroeconomic data set will be used to compare several models, including ARIMA, Random Forest and XGBoost, for prediction purposes.

Therefore, the forecasts should not be assessed solely based on the prediction accuracy. The practical usefulness of the model is equally important when considering its robustness to different economic environments.

2.6 Challenges in Inflation Forecasting

Despite its importance in macroeconomic analysis, forecasting the rate of inflation continues to be very difficult. This is due to the interaction of different variables in an economy which affect the inflation rate. Monetary policy, the labour market, consumer behaviour, market expectations and external events are all factors that influence inflation. Thus, historical dependence and changing economic dynamics must be included in inflation models.

Structural breaks are one of the challenges in predicting inflation. These are changes in economic dynamics brought about by external shocks to the economy. External shocks can be financial crises, epidemics, geopolitics or policy changes. Under these circumstances, a model of inflation using past data would have a hard time predicting inflation.

This is reflected in COVID-19 pandemic, in addition to the inflation rise between 2022 and 2023. It has been said that the chain of supply disruptions, energy prices and geopolitics have caused inflation to shoot up, and these factors have led to substantial

variations in inflation, making prediction a more challenging.



Figure 3: UK Inflation Dynamics and Crisis Periods (2000–2024)

As shown in Figure 3, the inflation rate in the UK has acted in certain ways during different periods of economic shocks. This covers the current COVID-19 economic shock as well as the recent shock of inflation in 2022-23. The reason for the surge in the rate of inflation seen during the post-COVID period is attributed to supply chain issues, higher energy costs, and geo-political instability. This has made forecasts more difficult.

The second big challenge in the inflation forecasting exercise is the presence of non-linearities among macroeconomic variables. Other factors affecting consumers' sentiment, such as employment and economic growth, may also affect the effect of interest rates on the inflation rate.

When forecasting inflation, the issue of lags is another complicating factor. There are numerous macroeconomic indicators that have a lag effect on inflation. For instance, changes in bank interest rates had significant impact on the level of inflation after a few months. Likewise, unemployment or consumer behaviour might affect the dynamics of prices after some time. When one macroeconomic variable impact another with a time lag, which must be incorporated into the model.

Another challenge to overcome in the construction of forecasting models for inflation is exogenous shocks. Unexpected events or situations which are not included in the current data set are Exogenous shocks. Some examples of such events are oil shocks, wars, supply shortages or political unrest, which can make a huge difference to the economic climate.

One of the restrictions is the quality of data, and the need to note that macroeconomic data is subject to revisions after initial publication, which could impact forecast

accuracy. Although machine learning approaches have a higher potential for accurate predictions, they also have other problems.

Among other things, machine learning techniques may have issues with:

- Overfitting
- Reduced interpretability
- Instability across different validation windows

Further, machine learning models may not be able to provide a good estimate of large jumps in inflation, if they are relatively infrequent, not reflected in the historical data used to train the model.

2.7 Research Gap

Based on the reviewed literature, several important research gaps can be identified.

First, it is related to the widespread application of statistical-based methods for economic prediction analysis in which the models employed assume linearity in the relationships between the variables in the data set. Although these models are interpretable, they are inadequate in dealing with the complexity and non-linear nature of actual economic processes.

Secondly, although machine learning methods have shown greater accuracy, they are frequently evaluated using only accuracy metrics and not considering the reliability of these models through various economic scenarios.

Another limitation frequently mentioned in literature is the limited integration of forecasting models into practical decision support systems. Most studies deal with forecasting accuracy and represent model outputs in terms of numerical predictions or static visualisations. Relatively little effort has been spent on translating forecasts into interpretable risk categories that can inform decision-making by policymakers, analysts and business users.

Moreover, many studies assess forecasting performance with a single train-test split. Such an approach may not adequately capture the robustness of the model in different economic regimes, especially in periods of structural breaks and extreme volatility of inflation.

Finally, there is little research that combines inflation forecasting, multi-window validation, crisis-risk classification and interactive scenario simulation in one framework. This represents an important research gap that this study seeks to address.

2.8 Contribution to this Study

This study addresses the identified research gaps by developing an integrated forecasting and decision-support framework for economic crisis prediction.

The primary contributions of this research include:

- Comparative implementation of statistical and machine learning forecasting models.
- Applying modern feature engineering techniques such as lagged values, momentum measures, and interaction terms.
- Using multi-window validation methodology to gauge the robustness of models in varying economic environments.
- The formulation of a threshold-driven economic crisis risk model for interpreting inflation forecasts.
- The development of a user-friendly economic dashboard for scenario simulation.
- Incorporation of forecasting results in an interactive Streamlit-based simulation tool allowing users to assess future inflation risk under various macroeconomic scenarios.

In contrast to other research endeavours, this study is not restricted to the predictive modelling process; rather, it incorporates the forecasted outcomes into a decision-making context that can be applied in real life.

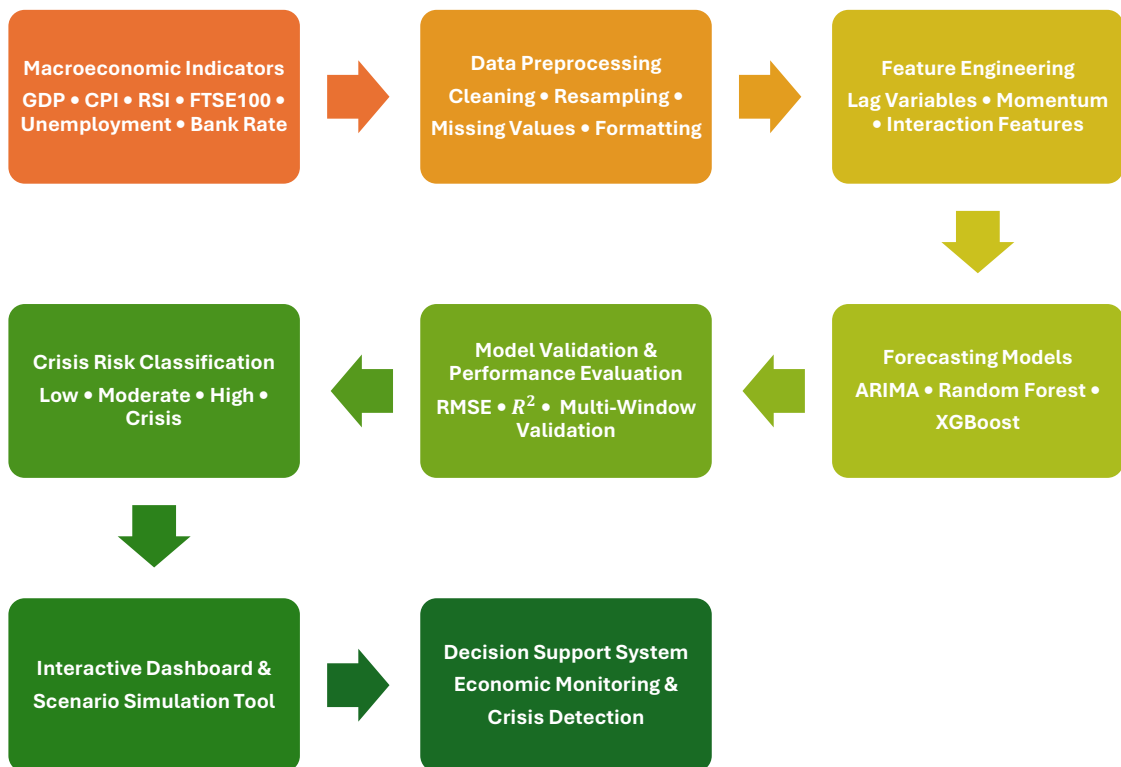


Figure 4: Proposed Forecasting and Decision-Support Framework

The entire framework that is proposed in this study is shown in Figure 4. The system consists of the utilisation of macroeconomic variables, preprocessing, feature extraction, prediction models, validation methods and crisis categorisation. This framework is not only used for the economic forecast; it allows to conduct economic monitoring and analysis in real time and with different scenarios.

2.9 Chapter Summary

This chapter reviewed literature related to inflation forecasting macroeconomic modelling, machine learning methods and economic crises detection. Moreover, advantages and disadvantages of classical statistical methods and machine learning models were outlined in this literature review.

From the literature, it was revealed that the economists still face challenges for forecasting inflation because of the presence of structural breaks, non-linearity, lags and external shocks which make the task difficult. The benefits of machine learning compared to classical statistical models in predicting economic phenomena are shown but the issues of transparency and instability of the machine learning model are critical.

Also, literature revealed some significant research gaps that need to be addressed by researchers.

The problem can be solved by developing and implementing a combined system that is a combination of comparative forecast framework with an economic crisis detection system based on thresholds and interactive dashboard. Next chapter discusses methodology used in this study.

This implies that just increasing the accuracy of the forecasts is not enough. Systems for forecasting the future need to focus on transparency, robustness and practical decision support capabilities.

The literature reviewed suggests that although predictive performance has been improved by machine learning techniques, namely Random Forest (Breiman, 2001) and XGBoost (Chen & Guestrin, 2016), forecasting performance is still vulnerable to structural breaks and changing economic conditions (Stock & Watson, 2008).

3. METHODOLOGY

3.1 Introduction

This chapter describes the methodology adopted for developing the inflation forecasting and economic crisis prediction framework proposed in this study. This approach integrates conventional statistical forecasting methods, machine learning algorithms, feature engineering approaches, and interactive decision-support systems in order to investigate the dynamics of inflation in the United Kingdom's economy.

In this study, the research method used is quantitative and data-oriented, using monthly data on macroeconomic indicators for the period from 2000 to 2024. Data modelling consists of different stages, which include data collection, preprocessing, exploratory data analysis, feature engineering, building of the forecasting model, testing of the model, and classification of crisis risks. The approach is geared towards producing meaningful economic risks from the forecasting model.

The forecasting process involves three modelling methods. The first method that was used is the ARIMA model as the base statistical model because of its extensive use in economic forecasting and time series modelling. Moreover, the Random Forest and XGBoost modelling methods were also considered to incorporate the non-linear relationship between economic factors and inflation behaviour.

For the purposes of adding more robustness to the process, several forms of validation processes were utilized, which include holdout validation at different economic times and walk-forward validation through time-series cross-validation methods. Furthermore, other feature engineering procedures like lagged variables, momentum variables, growth variables, and interaction variables were employed.

Furthermore, beyond predicting, the approach also encompasses the construction of a threshold-based system for detecting crises, where the inflation risks are classified according to the economic risks. The results from the forecast, alongside the classification of risks, are also embedded within Power BI and Streamlit dashboards to allow for scenario simulation and economic surveillance.

3.2 Research Design

This research study uses the methodology of quantitative research oriented at using various types of datasets to examine the correlation between macroeconomic variables and the development of predictive models aimed at forecasting economic crises in relation to predicting the inflation rate in the United Kingdom. The main goal of the research paper lies in the creation of predictive models capable of identifying the signs of inflationary tendencies and potential threats of economic crisis based on the trends identified by macroeconomic factors.

The research design is based on the methodology of the comparative model as the study utilizes a comparative model in which classical statistics-based methods and

machine learning are being evaluated in terms of their efficiency in predicting inflation rates depending on economic conditions. In such a manner, the comparative model uses ARIMA for forecasting inflation while random forest and XGBoost are used as examples of machine learning-based predictive models.

It is necessary to note that the main goal of this research design does not include any investigation of possible causal economic relationships among the studied variables. Instead, the purpose of the paper is to identify whether the use of macroeconomic data enhances the predictive performance of the models. This means that special attention should be paid to precision, robustness, and practicality of the models for further decision-making applications.

Time series research design is applied in the study as its goal includes analysis of the trend developed by macroeconomic data collected continuously during the period from 2000 to 2024. The monthly frequency was selected for macroeconomic data collection because this type of data provides enough information for analysis of the economic trend as well as short-term dynamics.

Additionally, this study is based on the application of experimental modelling using the technique of walking forward validation with several validation windows that correspond to periods of economic stability, disruptive economic conditions related to the spread of coronavirus pandemic, and rapidly growing inflation rate from 2022 to 2023.

In addition to predictive modelling techniques, the paper is oriented at decision making as its scope involves developing threshold-based framework as well as interactive dashboard system aimed at detecting crisis periods. This means that this study goes beyond the scope of forecasting papers as it deals with application of model outputs to generate scenarios of economic risks.

In summary, the research methodology involves the application of forecasting, comparative evaluation, validation, and simulation in a cohesive forecasting framework aimed at providing support to both economic analysis and decision-making.

3.3 Data Collection

For this research study, monthly macroeconomic indicators indicating the economic environment in the UK were used from the year 2000 to 2024. The data was acquired from credible sources of financial and economic information found in public domains to maintain uniformity, validity, and history of economic data.

This research focuses on macroeconomic variables which have been found associated with inflation behaviour and monetary policy analysis in past literature. These variables have been selected for the research due to their relevance in the literature and effects on the behaviour of inflation. The identified variables are the Consumer

Price Index (CPI), Gross Domestic Product (GDP), unemployment rate, Bank Rate, FTSE100 index, and Retail Sales Index (RSI).

Initially, the choice of the Consumer Price Index (CPI) as the indicator of inflation behaviour was made. However, stationarity tests showed that the initial CPI time series was non-stationary. Thus, the year-over-year (YoY) rate of change of CPI (Inflation_YoY) was calculated using CPI and taken as a predictor variable.

The following table summarises the macroeconomic variables used in this study.

Variable	Description	Purpose in this Study
CPI	Consumer Price Index	Used to derive Inflation_YoY
Inflation_YoY	Year-on-Year inflation rate	Target forecasting variable
GDP	Gross Domestic Product Growth	Economic growth indicator
Unemployment	UK unemployment rate	Labour market condition indicator
Bank Rate	Bank of England interest rate	Monetary policy indicator
FTSE100 Index	UK Stock Market Index	Financial market sentiment indicator
RSI	Retail Sales Index	Consumer spending behaviour indicator

Table 2: Summary for Macroeconomic variables

The macroeconomic data was collected from the following sources:

- Office for National Statistics (ONS)
- Bank of England
- Yahoo Finance
- Financial and economic open-data repositories

Since the variables were acquired from different sources and represented at different frequencies, they were standardized to a common monthly time series. Time indexing was conducted in terms of dates in Python.

The selected time horizon includes several important economic events and structural changes within the UK economy, including:

- the 2008 financial crisis recovery period,
- the COVID-19 pandemic,
- post-pandemic supply chain disruptions,
- and the 2022–2023 inflation surge.

By incorporating both stable and volatile economic environments into forecasting models, the ability of models to learn from inflation behaviour during different times becomes enhanced.

The data collected and standardized was then used for further processes including data pre-processing, exploratory data analysis, feature engineering, and forecasting model development as described in the following section.

3.4 Data Preprocessing

To enhance the quality and uniformity of the macroeconomic dataset for forecasting models, data preprocessing techniques were applied. Since the data was gathered from multiple sources, preprocessing is necessary to deal with problems arising from issues in the quality, uniformity, and compatibility of the datasets.

Data preprocessing was performed in two stages; the first stage involved standardizing the data to a similar frequency. This was due to different frequency levels of the data source used and varied formats of dates used. All variables were sorted in ascending order according to dates.

Missing values were handled through forward filling. This was important because macroeconomic variables do not tend to change abruptly. Therefore, using this method of handling missing values allowed me to maintain the continuity of the data without having abrupt jumps in data.

After conducting the integration of the datasets, the CPI variable was transformed into Inflation YoY which would be the dependent variable in the prediction model.

The year-on-year inflation calculation can be represented as:

$$Inflation_{YoY}_t = \left(\frac{CPI_t - CPI_{t-12}}{CPI_{t-12}} \right) \times 100$$

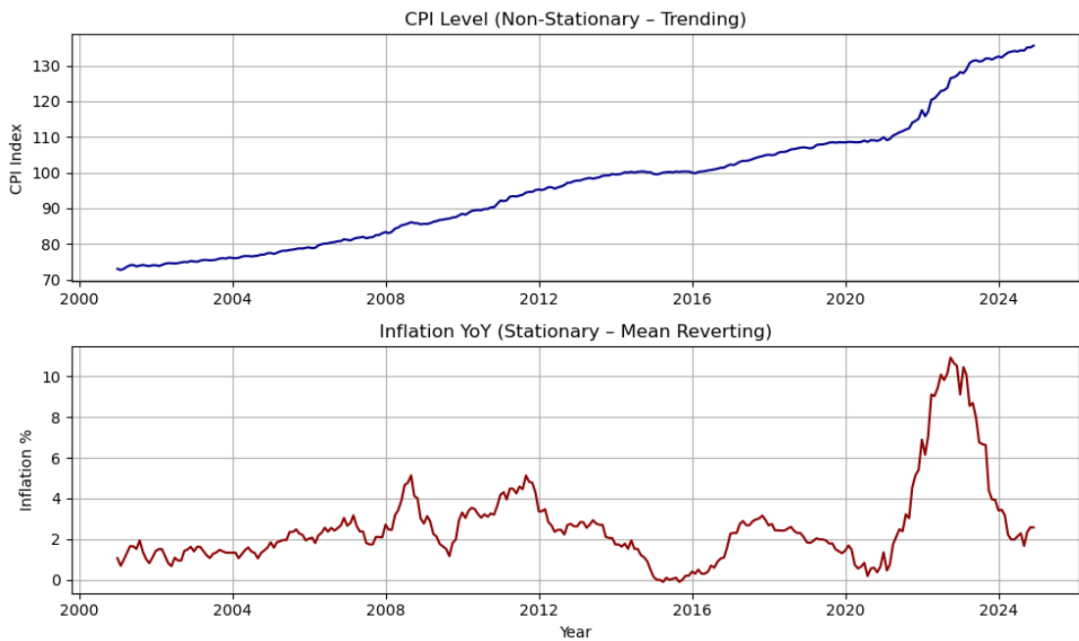


Figure 5: CPI to Inflation_YoY Transformation

Transformation was required since the original CPI variable showed non-stationarity behaviour during the data exploration and stationarity testing phase. Using Inflation_YoY would enable me to have stationary time series data to predict.

Stationarity of the CPI and Inflation_YoY was tested using the augmented Dickey-Fuller test. Although the CPI variable showed non-stationarity, Inflation YoY had higher stationarity levels after the transformation process. Stationarity is essential because models like ARIMA use time series data.

Data processing was done to some selected macroeconomic indicators whereby in case the change was required, percentage changes and difference were used to solve scale differences between macroeconomic indicators.

During the process of applying machine learning algorithms, feature scaling was done. Scaling of data was very important because the data I used includes variables that have been measured at various scales, which include interest rates, stock index and growth rates.

The standardisation process transformed variables into z-score representations using the following formula:

$$z = \frac{x - \mu}{\sigma}$$

where:

- x represents the original feature value
- μ represents the feature mean value
- σ represents the feature standard deviation

Finally, all processed data were compiled to a modelling dataset and used for exploratory data analysis, forecasting, and other procedures that will be explained in this research.

3.5 Feature Engineering

Feature engineering is done to improve prediction results. Feature engineering involves introducing new variables capable of capturing inflation persistence, delayed macroeconomic impact, and economic indicator interactions.

To capture both short and long-term behaviour impacts, lagged variables were introduced. Lagged variables help capture persistence features. The final dataset consisted of following lagged variables:

- Inflation_lag1
- Inflation_lag2
- Inflation_lag3
- Inflation_lag6
- Inflation_lag12

Moreover, change-based variables were introduced to capture short-term economic changes and macroeconomic momentum. The following change-based features were created:

- GDP_growth
- FTSE_returns
- Unemployment_change
- RSI_change

The growth rates in GDP and FTSE return were calculated through percentage change method, while the unemployment and retail sales changes were calculated through differencing.

In order to capture the momentum and direction behind inflation, indicators of inflation momentum were constructed. The following momentum features were incorporated:

- Inflation_momentum_3
- Inflation_momentum_6

These variables were calculated as the difference between current inflation and previous inflation values over 3-month and 6-month intervals respectively.

Interaction variables were also introduced to capture combined macroeconomic effects between economic indicators.

The following interaction variables were included:

- Rate_Unemployment
- Market_Sentiment



Figure 6: Feature Engineering Framework

Feature	Description
Inflation_lag1	Previous month inflation
GDP_growth	Monthly GDP growth
Market_Sentiment	FTSE $\times$ RSI interaction

Table 3: Engineered Features and Purpose

After feature engineering, the final modelling dataset contained both original macroeconomic indicators and transformed features, which were subsequently used

for exploratory data analysis, model implementation, and validation procedures discussed in the following sections.

3.6 Exploratory Data Analysis Methodology

Exploratory Data Analysis (EDA) was conducted to understand the behaviour, distribution, and relationships between macroeconomic variables before implementing the forecasting models. The aim of this step was to uncover trends in underlying inflation, structural shifts in the dataset, stationarity properties, and correlation between macroeconomic data and inflationary trends.



Figure 7: Exploratory Data Analysis Workflow

The initial step in exploratory data analysis consisted in visual inspection of trends in macroeconomic variables for the period of 2000-2024. Line graphs and time series graphs were utilized to study inflation trends, economic cycles, volatility, and major economic disruptions. Specifically, emphasis was placed on those time intervals corresponding to major structural shifts in the economy, such as the period of the coronavirus pandemic and the spike in inflation witnessed between 2022-2023.

For clarity, the moments of crisis in inflation were emphasized in certain visualizations in order to assess how the variable reacted to economic turmoil and what difficulties may arise when trying to forecast large-scale increases in inflation.

Stationarity testing formed a key part of the exploratory analysis because time series forecasting models such as ARIMA assume stationary data. The Augmented Dickey-Fuller (ADF) test was applied to both CPI and Inflation_YoY series to evaluate whether the statistical properties of the series remained stable over time.

The ADF hypothesis testing framework can be summarised as:

- Null Hypothesis (H_0): The series is non-stationary.
- Alternative Hypothesis (H_1): The series is stationary.

The results showed that the raw CPI series exhibited strong non-stationary behaviour, while the transformed Inflation_YoY series demonstrated improved stationarity characteristics. Thus, Inflation_YoY was chosen to be the target variable.

Furthermore, lagged relationships between macroeconomic variables and inflation were investigated through the use of correlation and lag analysis. Given that macroeconomic factors have time lags in affecting the inflation rate, lagged relationship heat maps and correlations were used to identify strength and direction of relationships across multiple lag periods.

Lag analysis was particularly important for:

- identifying inflation persistence,
- understanding delayed policy effects,
- and supporting feature engineering decisions.

For comparing variables that were recorded using different scales, visualizations using z-score normalization were created for a selection of macroeconomic variables. Graphs standardized in this way made it easier to compare relative trends in inflation, unemployment, Bank Rate, and financial variables in terms of a single scale.

Additionally, distributions were examined by way of histograms and density curves for selected macroeconomic variables. These allowed for the examination of variance, volatility trends, and abnormal intervals of inflation.

The exploratory data analysis stage provided key information about the nature and characteristics of the data set before implementing any form of prediction. These results guided the decision-making process for data preparation, selection of target variables, and modelling procedures detailed in the next section.

3.7 Forecasting Models

The analysis in this paper is based on a comparative forecasting methodology which combines both the conventional statistical methodology and machine learning techniques.

Three forecasting models were implemented in this study:

- ARIMA
- Random Forest
- XGBoost

ARIMA was chosen as the benchmark model based on its prevalence in economic forecasting. Random Forest and XGBoost were chosen as models capable of modelling nonlinear relationships and interactions between macroeconomic variables.

The models used for forecasting were built using the macroeconomic data set that was prepared in the earlier sections. The target variable considered in this analysis was Inflation_YoY, while the predictors consisted of macroeconomic indicators. To preserve the temporal structure of the data, the dataset was divided into time-based training and testing subsets.

The following subsections present the implementation and theory behind each forecast model.

3.7.1 ARIMA Model

The Autoregressive Integrated Moving Average (ARIMA) model was implemented as the baseline statistical forecasting approach in this study.

The ARIMA model combines three components:

- Autoregressive (AR)
- Integrated (I)
- Moving Average (MA)

The general representation of the ARIMA model is:

$$ARIMA(p, d, q)$$

Where:

- p represents the autoregressive order,
- d represents the differencing order,
- q represents the moving average order.

Stationarity tests and transformation processes were carried out before implementing the ARIMA model due to its assumption of a stationary process. Since the raw CPI series was found to be non-stationary, the transformed Inflation_YoY series was used for forecasting.

In this study, ARIMA served as a benchmark model against which machine learning models were compared.

3.7.2 Random Forest

Random Forest was implemented as the primary machine learning forecasting model within this study. Random Forest builds a collection of decision trees and their outputs are averaged together to produce a prediction.

The reasons for selecting Random Forest were because it can handle non-linearities in, and interactions between macroeconomic variables. The model was trained using:

- original macroeconomic variables,
- lag-based features,
- momentum indicators,
- growth variables,
- and interaction features.

3.7.3 XGBoost

Extreme Gradient Boosting (XGBoost) served as an advanced ensemble machine learning technique for the purpose of comparing and analysing the forecasts. The basis of XGBoost lies in gradient boosting, where models are learned one after another in order to improve upon the errors made by previous models.

XGBoost was selected in this study because of its:

- high predictive capability,

- regularisation mechanisms,
- computational efficiency,
- and ability to model non-linear relationships.

The model was trained using the same engineered feature set used for Random Forest to ensure fair comparison between machine learning approaches.

3.8 Model Evaluation Metrics

To evaluate the effectiveness of the forecasted results, several statistical metrics were used in this research. Since there is no one metric which can evaluate forecasting performance completely, both error and goodness-of-fit metrics were used to compare the models for different validation periods.

The forecasting models were evaluated primarily using:

- Root Mean Square Error (RMSE),
- Mean Absolute Error (MAE),
- and the coefficient of determination (R^2).

These metrics were selected because they are widely used in economic forecasting literature and provide complementary perspectives regarding model accuracy and predictive reliability.

Root Mean Square Error (RMSE)

Root Mean Square Error (RMSE) was used as the primary evaluation metric in this study. RMSE measures the average magnitude of prediction errors by calculating the square root of the average squared difference between actual and predicted values.

The RMSE formula is given as:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Where:

- y_i represents the actual inflation value,
- $\hat{y}_i$ represents the predicted inflation value,
- n represents the number of observations.

Lower RMSE results indicate better forecast accuracy since they mean lower discrepancies between actual and forecasted levels of inflation.

Mean Absolute Error (MAE)

Mean Absolute Error (MAE) measures the average absolute difference between predicted and actual values. Unlike RMSE, MAE does not square the forecasting errors and therefore provides a more direct interpretation of average prediction deviation.

The MAE formula is represented as:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

MAE was included to complement RMSE and provide additional understanding of forecasting consistency across validation periods.

Coefficient of Determination (R^2)

The coefficient of determination (R^2) was used to evaluate how effectively the forecasting models explained variation within the inflation data.

The (R^2) formula is defined as:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

Where:

- y_i represents actual values,
- $\hat{y}_i$ represents predicted values,
- $\bar{y}$ represents the mean of actual values.

The higher the value of R^2 , the greater is the explanatory power of the forecasting model. But since there is usually a break or volatility in the economic data used for forecasts, R^2 values were considered along with RMSE and MAE.

Importance of Multi-Metric Evaluation

The use of several evaluation measures was especially critical since there are stable times and highly volatile times when forecasting inflation. Some models can have good performance at one time and perform poorly during disruptive periods.

Therefore, evaluating forecasting performance using multiple metrics allowed for:

- more balanced comparison between models,
- assessment of robustness across validation windows,
- and identification of model limitations during extreme economic conditions.

The calculated evaluation metrics were subsequently used to compare model performances across different forecasting periods and validation strategies discussed in the following sections.

3.9 Validation Strategy

Model validation was conducted to evaluate forecasting robustness and generalisation capability across different economic environments. Since changes in the behaviour of the inflation process occur significantly throughout time because of changes in the

economic structure, changes in economic policies, and exogenous factors, relying on one train-test split could give erroneous results regarding the model performance.

The validation framework consisted of:

- holdout validation across multiple economic time-periods,
- multi-window validation,
- and walk-forward validation using time series cross-validation techniques.

Holdout Validation

In this study, the first method of validation that was utilized was holdout validation. In time series prediction, holdout validation involves using past observations for training and future observations for testing while maintaining the sequential nature of observations.

There were three different validation periods utilized in this study.

Validation Window 1

The first validation window used:

- Training Period: 2000–2018
- Testing Period: 2019–2022

This validation period was designed to evaluate model performance during the transition into highly volatile inflation conditions, including the COVID-19 pandemic and the early stages of the inflation surge.

Validation Window 2

The second validation window used:

- Training Period: 2000–2022
- Testing Period: 2023–2024

This validation period evaluated model behaviour during more recent economic conditions following the major inflation spike observed between 2022 and 2023.

Validation Window 3

The third validation window used:

- Training Period: 2000–2019
- Testing Period: 2020–2024

In the present validation window, emphasis was placed on forecast accuracy during the crisis period where the time periods for COVID-19 disruption and post-COVID-19 inflation were added in the test data set.

Walk-Forward Validation

In addition to the traditional holdout validation approach, walk-forward validation was also applied using the TimeSeriesSplit method. This approach is well suited for time series forecasting, as it reflects how forecasting is carried out in real-world situations. Instead of training the model on a fixed dataset, the training data is gradually expanded over time, allowing the model to make predictions using information that would realistically have been available at that point.

Under this approach:

1. The model was first trained using an initial set of historical observations.
2. Generate forecasts for the following time period.
3. The training dataset was expanded to include the newly available observations.
4. The process repeats iteratively across the dataset.

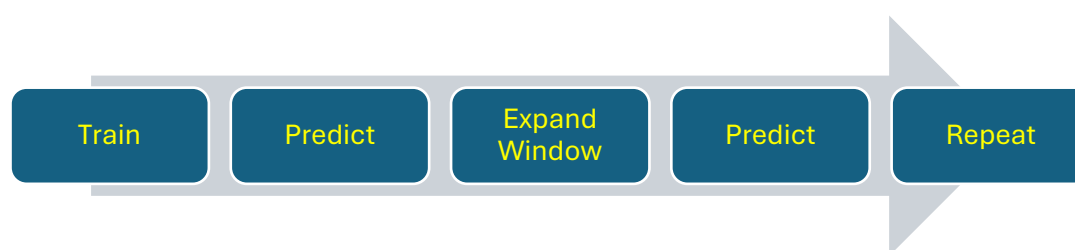


Figure 8: Walk-Forward Validation Process

The adopted validation strategy provided a comprehensive way to assess model reliability, forecasting stability, and practical usefulness within real-world economic forecasting scenarios.

3.10 Crisis Detection Framework

Alongside inflation forecasting, this study also proposes a crisis detection framework aimed at converting forecasted inflation values into clear and interpretable economic risk categories. The framework was developed to make forecasting results more meaningful from a practical decision-making perspective rather than presenting only numerical predictions.

Using this framework, predicted inflation values were grouped into four distinct economic risk categories:

- Low Risk
- Moderate Risk
- High Risk
- Crisis Level

The inflation risk thresholds used in this study are summarised below:

Predicted Inflation Level	Risk Classification
Inflation < 3%	Low Risk

Inflation between 3% and 5%	Moderate Risk
Inflation between 5% and 8%	High Risk
Inflation $\geq$ 8%	Crisis Level

Table 4: Risk Classification based on Inflation Level

In addition, the framework supports future economic scenario simulation. Users are able to modify macroeconomic variables such as GDP, unemployment, interest rates, and market-related indicators to examine how changing economic conditions may affect future inflation risk levels.

Although the threshold-based approach simplifies complex macroeconomic behaviour into predefined categories, it provides an interpretable method for translating forecasting outputs into economic risk insights.

3.11 Dashboard and Simulation Tool

To enhance the practical usefulness of the forecasting framework, this study also developed interactive dashboard systems alongside a simulation-based decision-support tool. These systems were designed to help users visualise inflation forecasts, track changing economic risk levels, and explore possible future macroeconomic scenarios in a more intuitive and user-friendly way.

Two separate dashboard platforms were developed during this study:

- Power BI
- Streamlit

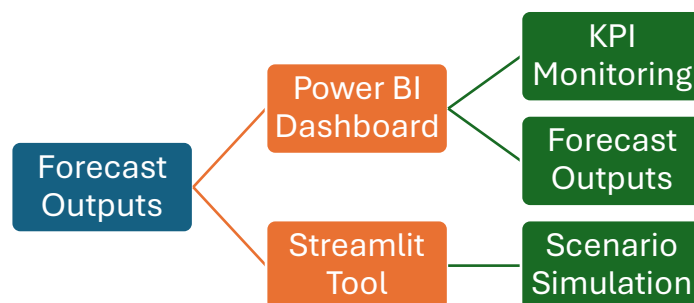


Figure 9: Exploratory Data Analysis Workflow

Power BI Dashboard

Power BI was initially chosen due to its strong business intelligence features, interactive visualisation capabilities, and suitability for executive-style reporting and analytical environments.

Streamlit Simulation Tool

Streamlit was implemented as an interactive simulation platform to extend the forecasting framework beyond traditional static dashboard visualisations.

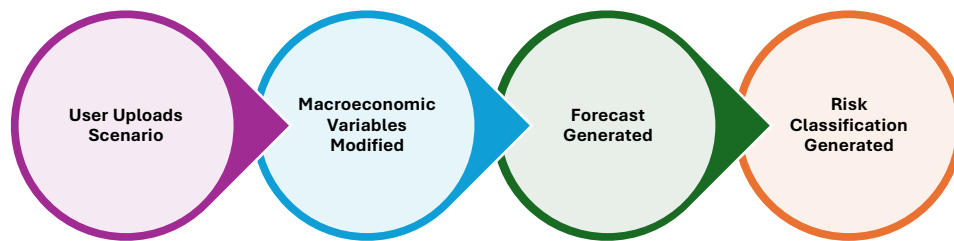


Figure 10: Simulation Workflow

Integration with Forecasting Framework

Both dashboard platforms were integrated with the outputs generated from the forecasting pipeline. Prediction datasets, confidence interval estimates, feature importance results, and crisis classification outputs produced during the modelling stage were exported and incorporated into the dashboard environments for visualisation and analysis.

The dashboard and simulation framework broaden the contribution of this research beyond conventional forecasting studies by providing an interactive decision-support environment that translates complex forecasting outputs into more accessible and interpretable economic insights.

The dashboard systems offer a practical and user-friendly interface that enables users to interact with forecasting outputs and examine potential future economic scenarios within a business intelligence setting.

3.12 Chapter Summary

This chapter outlined the methodology used to develop the inflation forecasting and economic crisis prediction framework proposed in this study. The selected approach combined macroeconomic data collection, preprocessing, feature engineering, forecasting model development, validation procedures, and interactive dashboard implementation.

The datasets used are UK macroeconomic data from year 2000-2024 where CPI was transformed into the Inflation_YoY target variable following stationarity assessment. Preprocessing and feature engineering techniques were applied to capture inflation persistence, lagged macroeconomic effects, and interactions between economic indicators.

Three forecasting approaches were implemented and evaluated within the study: ARIMA, Random Forest, and XGBoost. Multiple evaluation metrics and validation strategies were introduced to assess forecasting accuracy, model stability, and robustness across different economic periods.

Beyond conventional forecasting, the methodology also incorporated a threshold-based crisis detection framework together with interactive dashboard systems developed using Power BI and Streamlit.

The methodology established a comprehensive analytical framework capable of supporting both inflation prediction and economic risk interpretation.

4. EXPLORATORY DATA ANALYSIS

4.1 Introduction

This chapter presents the exploratory data analysis (EDA) conducted on the macroeconomic dataset used in this study. The purpose of this analysis is to examine the statistical behaviour, temporal patterns and relationships between macroeconomic indicators and inflation prior to the implementation of the forecasting models.

The analysis focuses on identifying:

- long-term inflation trends,
- distribution characteristics of macroeconomic variables,
- correlations between economic indicators,
- lagged relationships,
- and behavioural changes during periods of economic disruption.

Special attention is paid to structural changes related to the COVID-19 pandemic and the inflation surge between 2022 and 2023.

The analysis is a tool to understand the inflation persistence, macroeconomics relationships, forecasting problems and to justify the modelling choices we made in the previous chapter.

The following sections present the exploratory analysis of the selected macroeconomic variables.

4.2 Inflation Trend Analysis

Understanding historical inflation behaviour is essential before implementing forecasting models. This is because inflation patterns are often shaped by changing macroeconomic conditions, policy interventions and external economic shocks. Trend analysis can be useful in identifying periods of stability, volatility and structural change in the inflation series.

Figure 11 depicts UK year-on-year inflation (Inflation_YoY) movement between 2000 and 2024.



Figure 11: UK Inflation Trend (2000–2024)

The inflation series shows different economic regimes during the observation period. In the early 2000s inflation was relatively stable, moving in a moderate range around the target inflation range set by the Bank of England. However, temporary increases in inflation can be seen over periods of economic stress and financial instability.

During the global financial crisis there was a clear period of volatility in the recovery period after 2008. While inflation fluctuated during this period, inflation behaviour was still relatively moderate compared to later years.

The most important structural change within the data set was during the COVID-19 pandemic and the following post-pandemic recovery period. Initial inflation fell during the early pandemic period on account of depressed economic activity and lower consumer demand. However, from 2021 onward, inflation increased rapidly and reached historically elevated levels during 2022–2023.

The sharp increase in inflation during this period is likely to be related to a number of interacting factors including:

- supply chain disruptions,
- rising energy prices,
- labour market volatility,
- geopolitical tensions,
- and post pandemic recovery.

The trend analysis also reveals the existence of volatility clustering in the inflation series, with periods of relative stability of inflation, followed by sudden episodes of high volatility. This kind of behaviour increases the complexity of forecasting and reduces the effectiveness of models that rely heavily on stable historical relationships.

In summary, the analysis of the inflation trend demonstrates that the behaviour of the UK inflation from 2000 to 2024 involves relatively stable and extremely volatile

economic periods, which makes it a suitable dataset to test the robustness of the forecasting under different macroeconomic environments.

4.3 Stationarity Analysis

Stationarity testing was performed to determine if the statistical characteristics of the inflation series had changed over time. The stationarity of a series is a critical component of many time-series forecasting models since using non-stationary data can result in inaccurate forecasts and model instability. Typically, variables like inflation and price indexes used in macroeconomic forecasting exhibit long-run trends, varying variances, and/or structural breaks due to economic events and/or policy decisions. Therefore, determining and correcting for non-stationarity was a vital step prior to developing time-series forecasting models.

In our analysis of stationarity, we compared:

- the original CPI series,
- and the transformed Inflation_YoY series.

As shown in Figure 12, the original CPI series exhibited a significant upward trend throughout the entire study period.

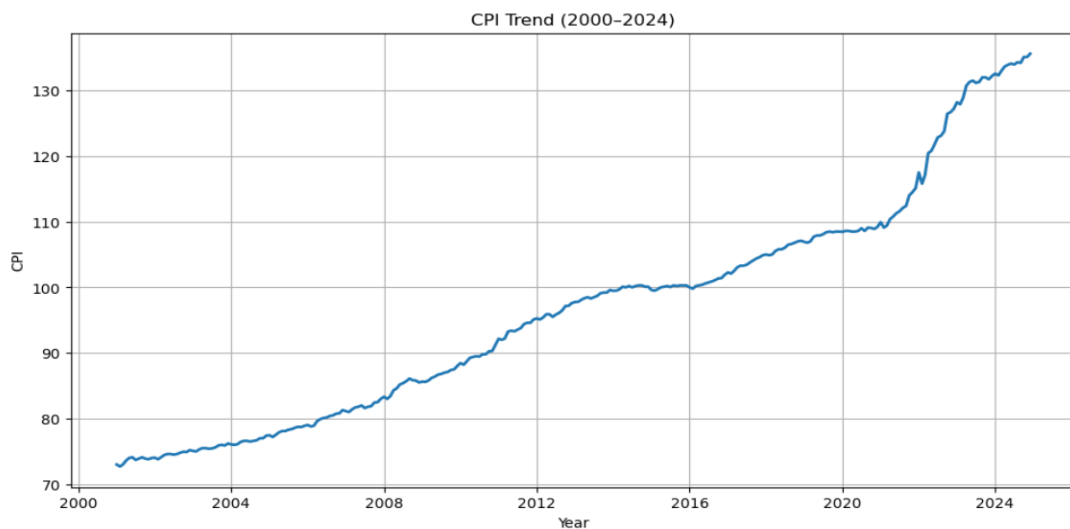


Figure 12: Consumer Price Index (CPI) Trend (2000–2024)

This trend is indicative of long-run non-stationary behaviour. Instead of oscillating about a constant mean value, the CPI series continued to increase during high inflation periods. This type of trend behaviour is inconsistent with the stationarity assumptions underlying models such as ARIMA.

The ADF test framework is based on two hypotheses:

- Null Hypothesis (H0): The series is non-stationary.
- Alternative Hypothesis (H1): The series is stationary.

Based on the ADF test results, the CPI series failed to reject the null hypothesis of non-stationarity, confirming the presence of strong trend behaviour within the data. To fix this issue, CPI was transformed into a year-on-year inflation measure (Inflation_YoY) by calculating the percentage change over a 12-month period.

Compared to the original CPI series, the transformed Inflation_YoY variable shows much better stationarity characteristics. While some volatility and structural changes are still present, the transformed series fluctuates around a more stable range instead of continuously increasing.

The stationarity analysis provided strong reasons for choosing Inflation_YoY as the target forecasting variable throughout this study and confirmed that transforming CPI into Inflation_YoY improved the dataset's statistical suitability for inflation forecasting and economic modelling.

4.4 Distribution Analysis

Distribution analysis was done to examine the spread, concentration, and variability of the macroeconomic variables used in this study. Understanding how variables are distributed is important because forecasting performance can be influenced by skewness, volatility, outliers, and irregular data behaviour.

The analysis focused on selected variables including:

- Inflation_YoY,
- GDP_growth,
- Unemployment,
- Bank_Rate
- FTSE100 returns,
- and Retail Sales Index (RSI) changes.

Figure 13 presents the distribution of the Inflation_YoY variable.

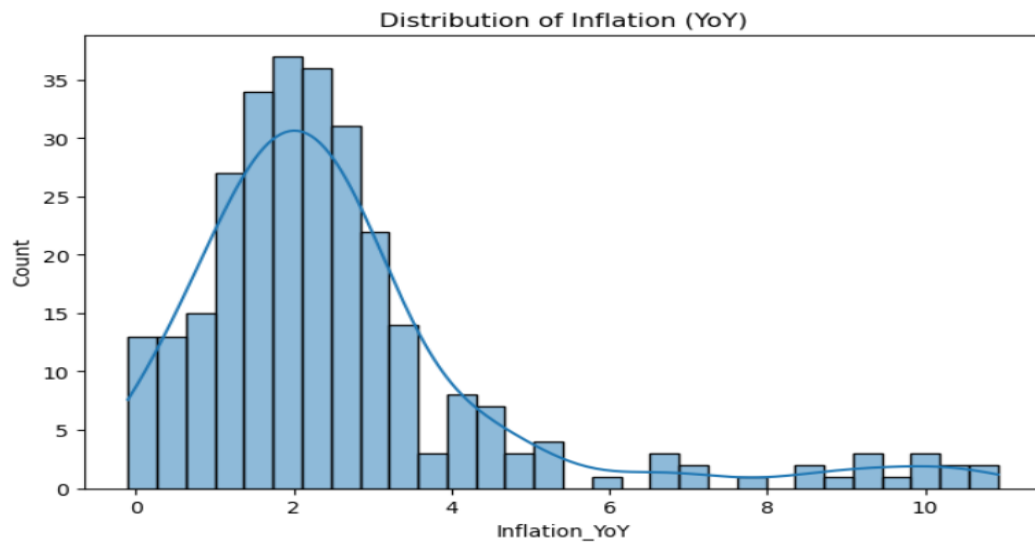


Figure 13: Distribution of Inflation_YoY

The distribution of Inflation_YoY shows that most observations fall within moderate inflation ranges. However, a smaller number of observations reflect periods of unusually high inflation. This creates a positively skewed distribution because of the extreme inflation spike observed during 2022 and 2023.

The presence of skewness and extreme values is important for forecasting because:

- forecasting models may struggle to generalize during rare inflationary events,
- prediction errors tend to rise during extreme economic conditions,
- models trained mainly on stable periods might underestimate sudden inflation surges.

The distribution analysis also indicates that inflation behaviour throughout the study period is not perfectly normally distributed. Such patterns are common in macroeconomic datasets since economic crises and policy shifts can create abnormal deviations from historical trends.

Distribution analysis was also performed for other macroeconomic indicators.

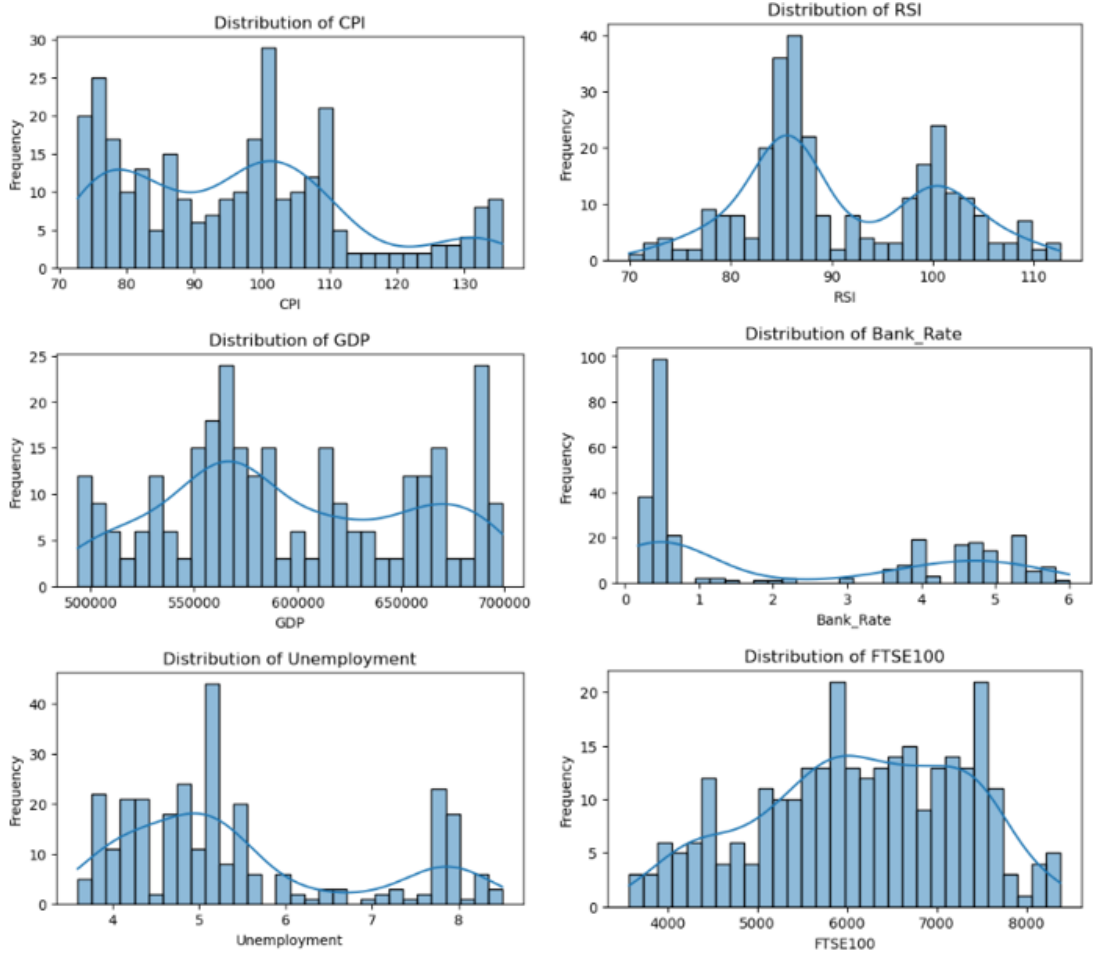


Figure 14: Distribution of Selected Macroeconomic Variables

Figure 14 presents the distributions of selected macroeconomic variables used within the forecasting framework.

The distributions of macroeconomic variables show different levels of volatility and dispersion across the dataset. For example, FTSE100 returns and inflation-related indicators show greater variability compared to more stable indicators like unemployment and the Bank Rate.

The differences in scale and variability highlighted the need for preprocessing and feature scaling, which were described in Chapter 3. Standardization was particularly crucial because machine learning models may become sensitive to variables measured across very different ranges.

The skewed inflation distribution suggests that rare inflation spikes may be difficult for forecasting models to learn from.

4.5 Correlation Analysis

Correlation analysis was done to explore the link between the inflation and the selected macroeconomic variables employed in the present study. Understanding

correlations between variables is vital as inflation behaviour is affected by interrelated economic situations, rather than isolated indicators.

The analysis was carried out to determine:

- the strength of the relationships between variables,
- the direction of these relationships,
- and the extent to which macroeconomic indicators can improve the performance of inflation forecasting.

A correlation matrix and heatmap visualisation were created using a Pearson correlation coefficient.

Figure 15 presents the correlation heatmap for the selected macroeconomic variables and engineered features.

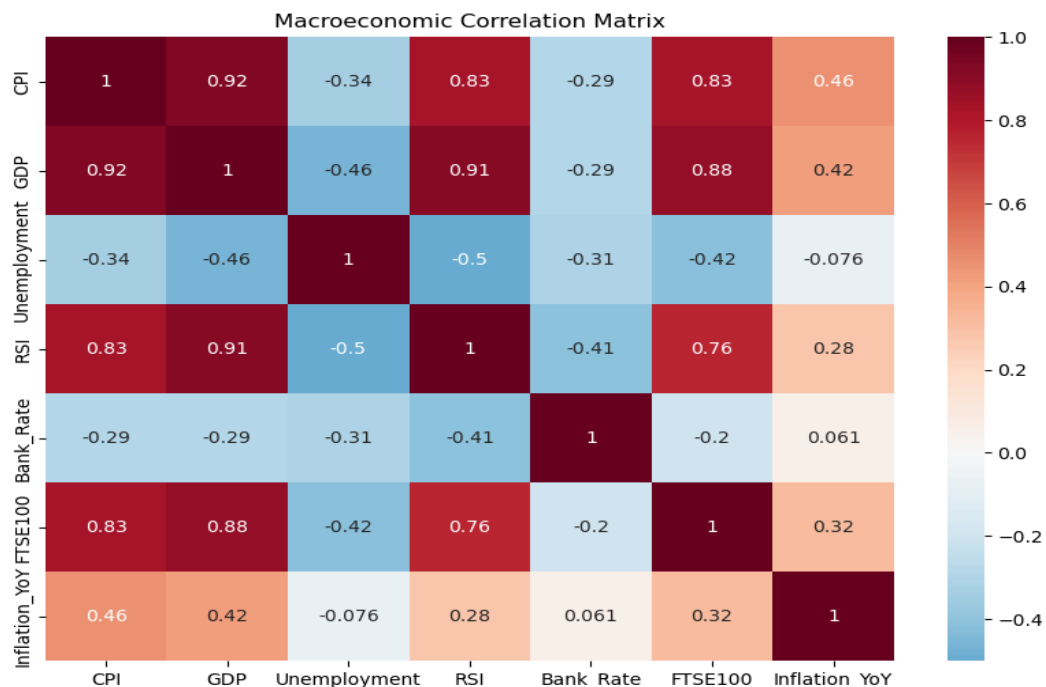


Figure 15: Correlation Heatmap of Macroeconomic Variables

Numerous significant relationships within the dataset are revealed by the correlation analysis. Inflation persistence over time was demonstrated by the strong positive correlations found between Inflation_YoY and its lag factors. This implies that historical inflation levels offer crucial information for predicting inflation behaviour in the future.

Additionally, macroeconomic factors like the bank rate show significant correlations with inflation. This observation is consistent with economic theory, which links monetary policy changes to efforts to control inflation and stabilise the economy.

The data also shows that, in comparison to their original factors, some manufactured traits show higher associations with inflation. Specifically:

- Lag-based variables,
- Momentum indicators,
- and interaction features

seem to do a better job of capturing inflation trends than just using macroeconomic indicators.

The heatmap also highlights the presence of correlations among predictor variables themselves. Because economic indicators frequently interact concurrently within the larger economy, such interactions are typical within macroeconomic datasets.

It's crucial to remember that correlation does not imply causation. Finding patterns and linkages pertinent to forecasting performance was the goal of this investigation rather than establishing causal economic relationships.

Hence, the correlation analysis supported all the feature engineering and forecasting techniques used in this work and offered significant insight into the macroeconomic dataset's structure.

4.6 Lag Relationship Analysis

Lag relationship analysis was conducted to investigate delayed interactions between macroeconomic indicators and inflation behaviour. In macroeconomic systems, changes in economic conditions and policy can have delayed consequences on inflation rates. Rather, inflation is influenced by numerous factors after a particular amount of time.

Lag correlation analysis was used over several time periods to examine these delayed effects. To ascertain whether historical economic variable values influenced future inflation movements, correlations between inflation and a few macroeconomic indicators were assessed using various lag intervals.

Figure 16 presents the lag correlation heatmap generated for selected macroeconomic indicators and inflation.

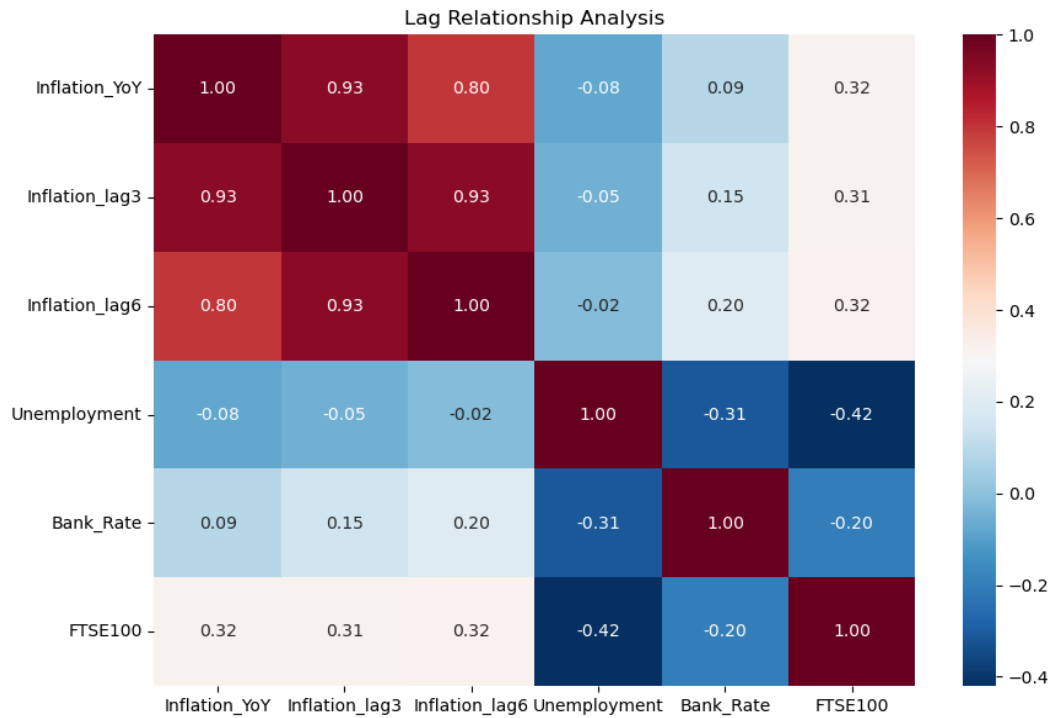


Figure 16: Lag Correlation Heatmap

The lag analysis reveals that inflation demonstrates strong temporal persistence across multiple lag periods. Specifically, lagged inflation variables show consistently high correlations with future inflation levels, suggesting that past inflation behaviour has a significant impact on later inflation movements.

This finding supports the inclusion of lag-based features such as:

- Inflation_lag1,
- Inflation_lag2,
- Inflation_lag3,
- Inflation_lag6,
- and Inflation_lag12

within the forecasting framework.

Additionally, the data indicates that some macroeconomic factors have delayed rather than immediate effects on inflation. For instance, it could take many months for changes in the bank rate and unemployment rate to have an effect on inflation behaviour.

The lag heatmap further emphasises how variable associations change over time. While certain markers grow more significant over longer forecasting horizons, others show greater predictive importance at shorter lag periods.

These results supported the employment of machine learning models that can understand intricate temporal correlations in macroeconomic data and further validated the feature engineering approach used in this work.

4.7 Crisis Period Behaviour Analysis

This study examined inflation behaviour throughout significant crisis-related times in the UK economy to look at these consequences, with a focus on:

- the COVID-19 epidemic,
- and the 2022–2023 spike in inflation.

Figure 17 illustrates UK inflation behaviour during major crisis periods across the different timelines.

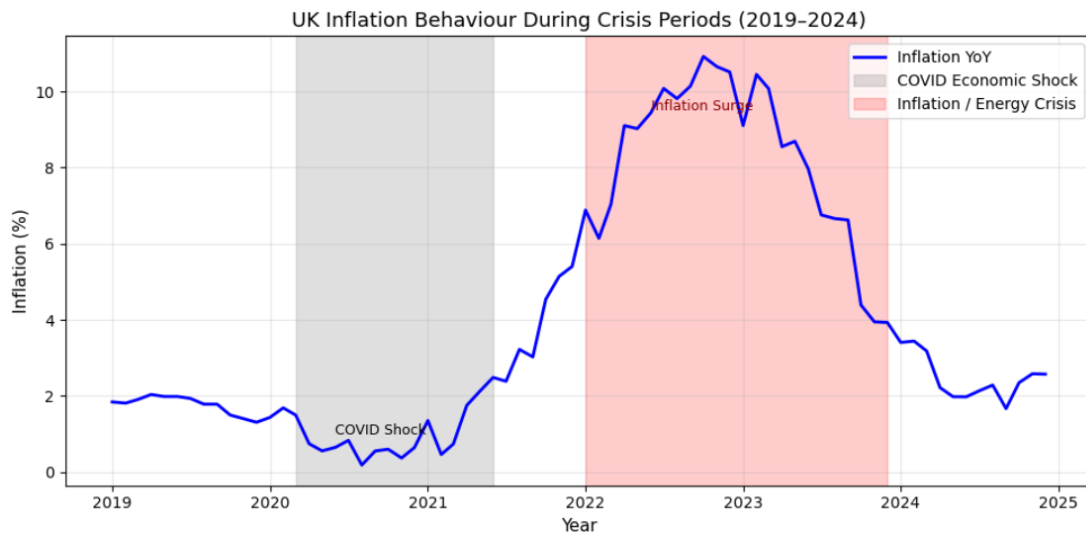


Figure 17: UK Inflation Dynamics During Crisis Periods

The figure demonstrates that inflation behaviour changed significantly during periods associated with economic disruption. Inflation first decreased in the early phases of the COVID-19 pandemic as a result of decreased economic activity, decreased consumer demand, and general market instability.

But after the pandemic recovery period, inflation rose quickly, to historically high levels in 2022 and 2023. One of the most erratic times in the dataset is represented by this inflation spike.

Several interacting factors likely contributed to this behaviour, including:

- supply chain disruptions,
- rising global energy prices,
- labour shortages,
- post-pandemic demand recovery,
- and geopolitical instability.

The crisis-period analysis shows that the inflation series has structural breakdowns. When historical economic relationships drastically shift over time, this is known as a structural break, and it lowers the accuracy of models that were predominantly trained on past economic behaviour.

The approach also explains why forecasting models in this study had higher prediction errors during the spike in inflation in 2022–2023. Machine learning models may find it difficult to properly generalise in these situations since such high inflation episodes are rather uncommon in the historical dataset.

The crisis-period analysis supports a few significant findings from a forecasting standpoint:

- inflation dynamics are highly sensitive to external shocks,
- macroeconomic relationships are not always stable,
- and forecasting robustness must be assessed under various economic conditions.

The research of crisis-period behaviour shows that inflation forecasting becomes significantly more difficult during times of structural economic disturbance and emphasises the significance of forecasting techniques that are robustness-focused on macroeconomic prediction systems.

4.8 Key Insights from EDA

The statistical and behavioural features of the macroeconomic dataset utilised for inflation forecasting were significantly clarified by the exploratory data analysis carried out in this chapter. The analysis showed that both long-term macroeconomic trends and abrupt structural disruptions brought on by external economic shocks have an impact on inflation behaviour within the UK economy.

The unusual inflation movements caused by these structural disruptions were very different from the historical patterns seen in previous years. Further evidence of inflation's great temporal persistence came from correlation and lag analysis. The inclusion of lag-based characteristics in the forecasting framework was supported by the significant connections that lagged inflation indicators showed with future inflation behaviour.

Additionally, the exploratory research demonstrated the intricate and interrelated interactions between macroeconomic indices. distinct time periods and lag intervals showed distinct correlations between variables related to consumer activity, financial markets, labour market circumstances, and monetary policy and inflation.

With everything considered, the exploratory data analysis laid a solid analytical basis for the forecasting models used in the ensuing chapters and offered crucial insight into the difficulties of anticipating inflation behaviour under shifting macroeconomic circumstances.

4.9 Chapter Summary

This chapter presented the exploratory data analysis conducted on the macroeconomic dataset used for inflation forecasting and crisis prediction. The statistical behaviour,

historical trends, distributional features, and correlations between macroeconomic factors and inflation dynamics in the UK economy were all explored.

The CPI was converted into the Inflation_YoY as target variable when the chapter showed that the previous CPI series had strong non-stationary behaviour. Important information about inflation persistence, delayed macroeconomic effects, and the interdependence of economic indicators was revealed by stationarity analysis, distribution analysis, correlation analysis, and lag relationship analysis.

Lastly, the exploratory data analysis provided a solid analytical basis for the forecasting models created in the next chapter. The forecasting pipeline utilised throughout the study is covered in the following chapter, which also shows how the ARIMA, Random Forest, and XGBoost forecasting models are implemented.

5. MODEL IMPLEMENTATION, RESULTS AND VALIDATION

5.1 Introduction

This chapter presents the implementation, evaluation, and validation of the forecasting models developed for inflation prediction within this study. The chapter evaluates the efficacy of various modelling methodologies under various economic conditions by combining forecasting implementation procedures with comparative performance analysis and robustness assessment.

Three forecasting approaches were implemented and compared:

- ARIMA,
- Random Forest,
- XGBoost

The macroeconomic dataset processed was used to train the models, together with the engineering features that were previously discussed. Macroeconomic indicators and transformed features were employed as predictor factors, with Inflation_YoY serving as the target forecasting variable.

The purpose of the implementation procedure was to assess how well various modelling techniques captured the dynamics of inflation under various economic circumstances. The behaviour of models during times of structural economic disturbances, such as the COVID-19 pandemic and the 2022–2023 inflation surge, were taken into consideration.

The chapter additionally describes:

- model robustness,
- validation stability,
- forecasting consistency,
- and the ability of models to generalise across changing economic regimes.

To maintain the sequential character of economic time series data, training and testing datasets were arranged chronologically for all forecasting models. To increase robustness and lower the chance of information leakage, several forecasting windows and validation techniques were used.

The chapter further examines:

- comparative model performance,
- feature importance analysis,
- actual versus expected inflation behaviour,
- and forecasting constraints under crisis-related economic circumstances.

Each forecasting model's implementation and outcomes are covered in the subsequent sections, which are followed by a comparative assessment and robustness study across several validation environments.

5.2 ARIMA Implementation and Results

This study used the Autoregressive Integrated Moving Average (ARIMA) model as its baseline statistical forecasting model. Because of its extensive use in macroeconomic forecasting and its potential to represent temporal relationships using past observations in a time series, ARIMA was chosen.

Since the model mainly depends on the target variable's past behaviour rather than several external predictors, the ARIMA implementation only looked at the Inflation_YoY series. Stationarity analysis was done before deployment to make sure the model's forecasting assumptions were met.

As discussed in Chapter 4, the original CPI series exhibited strong non-stationary behaviour. Since the altered Inflation_YoY series showed better stationarity characteristics after transformation, it was utilised for ARIMA forecasting. To maintain the temporal structure of the time series data, the dataset was split chronologically into training and testing periods. Since random splitting can induce future knowledge leaking into the training process, it was avoided.

The training dataset's historical inflation observations were used to train the ARIMA model, which was then utilised to produce projections for unobserved future periods.

Figure 18 presents the ARIMA forecasting results compared with actual inflation values.

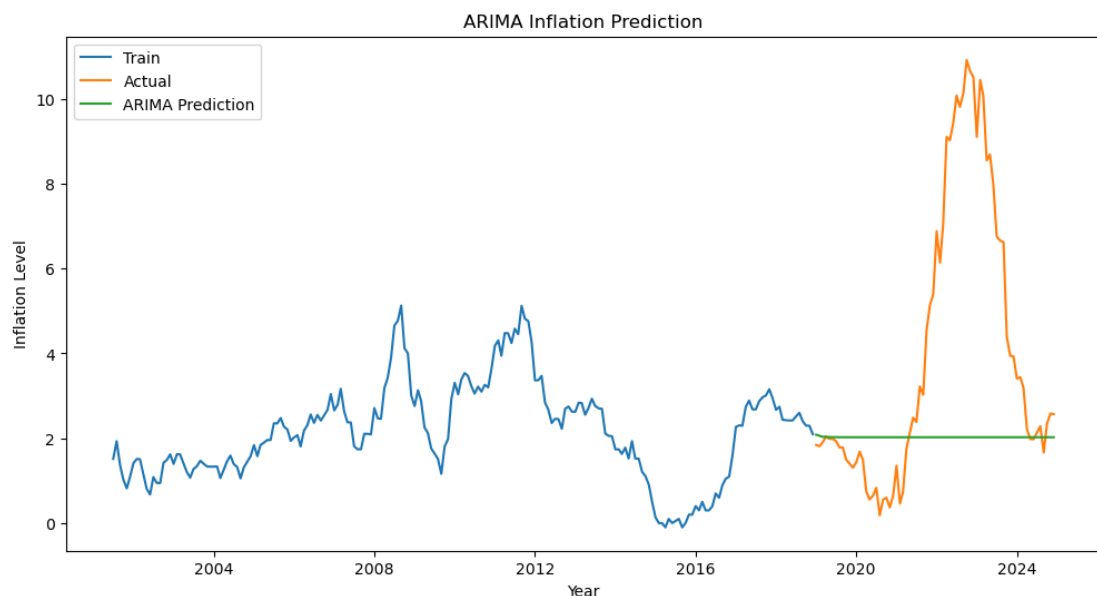


Figure 18: ARIMA Actual vs Predicted Inflation

The forecasting findings show that during comparatively stable economic times, the ARIMA model was able to accurately predict the long-term direction of inflation.

During times when inflation fluctuations remained moderate, the model maintained relatively consistent forecasts, as seen in Figure 18. However, the ARIMA model's predictions exhibited little reactivity to abrupt shifts in inflation behaviour and were mostly steady within a small range.

During the post-pandemic inflation spike that occurred between 2022 and 2023, significant forecasting problems were apparent. The extent and unpredictability of this structural shift were not captured by the ARIMA forecasts, despite the fact that real inflation climbed dramatically and surpassed 10%. Even though the observed inflation series changed quickly, the model provided comparatively flat projections and significantly underestimated the inflation spike.

This behaviour is indicative of several issues that are frequently connected to conventional statistical forecasting models. ARIMA's forecasting power is diminished when inflation dynamics are impacted by abrupt structural disruptions and non-linear macroeconomic interactions because it mostly depends on historical temporal patterns. Furthermore, the model is less useful in economic situations associated to crises since it presupposes comparatively stable historical linkages.

Measures like Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) were used to assess the ARIMA model's predicting ability.

Metric	Score
RMSE	3.88
MAE	2.616

Table 5: ARIMA Forecasting Performance Summary

The relatively high RMSE value indicates that the model struggled to accurately forecast periods associated with extreme inflation volatility. This was especially noticeable during the inflation spike of 2022–2023, when forecast errors significantly increased because of the model's incapacity to adjust to quickly shifting economic conditions.

Despite these drawbacks, ARIMA was crucial to this study's comparative forecasting approach. The model offered a statistical standard by which the machine learning models could be assessed. The comparison also demonstrated the drawbacks of using only conventional linear time series methods for inflation forecasting during times of structural economic instability.

From a larger forecasting perspective, the ARIMA results imply that more adaptable forecasting techniques that can capture non-linear correlations and changing macroeconomic interactions may be necessary to account for inflation behaviour during times of crisis.

The ARIMA implementation created a helpful standard for assessing the efficiency and resilience of the machine learning forecasting models covered in the ensuing parts.

5.3 Random Forest Implementation and Results

Random Forest was chosen as the primary machine learning forecasting model for this study because of its ability to capture nonlinear correlations and complicated interactions between macroeconomic indices. In contrast to ARIMA, which mainly depends on the target series, historical temporal behaviour, Random Forest simultaneously integrates several macroeconomic variables and engineering features during the forecasting process.

Random Forest model was trained using following components:

- lag-based inflation features,
- momentum indicators,
- growth variables,
- and interaction features developed during the feature engineering stage.

To retain the time series data's sequential structure and stop information leaking from subsequent observations, chronological train-test separation was implemented throughout implementation.

Several decision trees were created during training as part of the ensemble learning approach used to execute the model. In order to improve forecasting stability and lessen overfitting, final inflation estimates were generated by combining the results of different trees.

One of the major advantages of Random Forest within this study was its ability to model complex relationships between inflation and macroeconomic variables without requiring strict linear assumptions. As a result, the model was better able to adjust to shifting economic situations than conventional statistical forecasting techniques.

Figure 19 presents the Random Forest forecasting results compared with actual inflation values.

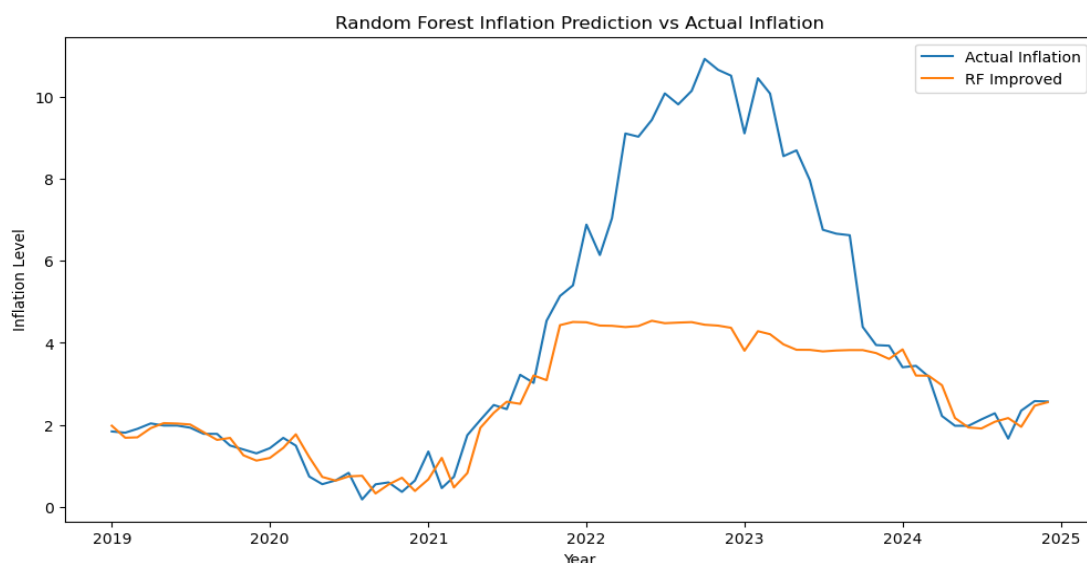


Figure 19: Random Forest Actual vs Predicted Inflation

The forecasting results show that, in comparison to the ARIMA baseline model, the Random Forest model significantly increased predictive ability. The model showed better reactivity to shifts in inflation behaviour over several economic periods and was able to track the general inflation trend more precisely.

Specifically, the Random Forest model had a greater capacity to capture variations linked to the post-pandemic inflation spike. The model generated projections that were far closer to actual inflation behaviour than ARIMA, even if some underestimating of peak inflation values persisted.

This enhanced performance is probably due to:

- the inclusion of engineered macroeconomic features,
- the ability of Random Forest to capture non-linear interactions,
- and the ensemble learning structure of the model.

The forecasting performance of the Random Forest model was evaluated using MAE and RMSE metrics.

Table 5.2 summarises the forecasting performance achieved by the Random Forest model.

Metric	Score
RMSE	≈ 2.64
MAE	≈ 1.55

Table 6: Random Forest Forecasting Performance Summary

The Random Forest model achieved the strongest overall forecasting performance among the implemented models.

The decrease in predicting error indicates that machine learning techniques were better at simulating inflation behaviour in intricate and unstable macroeconomic settings. The capacity of Random Forest to produce feature significance scores was another significant benefit. By pinpointing the macroeconomic factors that had the biggest impact on inflation forecasting performance, these scores added interpretability.

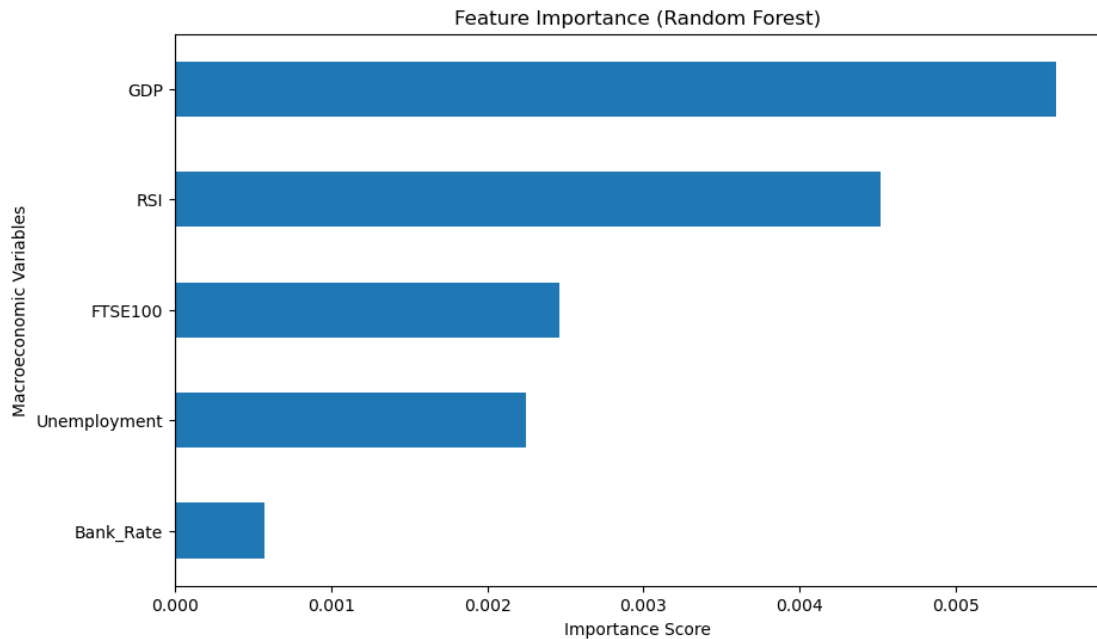


Figure 20: Random Forest Feature Importance

Figure 20 shows the feature importance analysis, which gives more information about the macroeconomic variables that contributed most significantly to the Random Forest model's inflation forecasting performance.

GDP had the greatest relevance score among the chosen variables, suggesting that total economic growth conditions had a significant impact on inflation behaviour over the course of the study. Additionally, the Retail Sales Index (RSI) demonstrated considerable predictive relevance, indicating a close relationship between inflation dynamics and consumer spending activity and retail market conditions.

The FTSE100 variable, which represents financial market conduct, also shown a significant impact to forecasting success. This suggests that in addition to conventional macroeconomic indicators, market sentiment and general economic confidence may have an impact on changes in inflation.

The relationship between labour market circumstances and inflationary pressure was reflected in the model's moderate predictive importance for unemployment. On the other hand, compared to other variables in the forecasting framework that was put into place, Bank Rate gave relatively lower relevance scores.

According to the feature importance analysis, the dataset's inflation behaviour was impacted by a mix of:

- economic growth conditions,
- consumer activity,
- financial market performance,
- and labour market behaviour.

These results also show that using a variety of macroeconomic variables greatly enhanced predicting ability as compared to depending only on past inflation trends.

From a more comprehensive forecasting point of view, the Random Forest implementation showed better capacity to capture intricate non-linear relationships between variables and greater adaptability to shifting macroeconomic situations. The model produced the best predicting performance in this investigation and continuously outperformed the ARIMA baseline during erratic economic times.

Therefore, the Random Forest model showed the value of machine learning techniques in macroeconomic forecasting applications and offered a reliable and understandable forecasting framework for inflation prediction.

5.4 XGBoost Implementation and Results

Extreme Gradient Boosting (XGBoost) was implemented as an advanced machine learning forecasting model to further evaluate the effectiveness of ensemble learning techniques within inflation forecasting applications.

Because of its robust prediction power, regularisation techniques, and capacity to represent intricate non-linear connections in macroeconomic datasets, XGBoost was chosen.

The XGBoost model was trained using the same methods as the Random Forest implementation:

- macroeconomic indicators,
- lag-based inflation features,
- momentum variables,
- growth indicators,
- and interaction features generated during feature engineering.

To maintain temporal consistency and stop future information leaking, chronological train-test splitting was upheld throughout implementation.

XGBoost develops trees sequentially by optimising residual prediction errors produced by earlier iterations, in contrast to Random Forest, which generates many separate decision trees. The model's ability to capture subtle relationships and increase forecasting accuracy across complex datasets is made possible by this iterative learning structure.

Figure 21 presents the XGBoost forecasting results compared with actual inflation values.

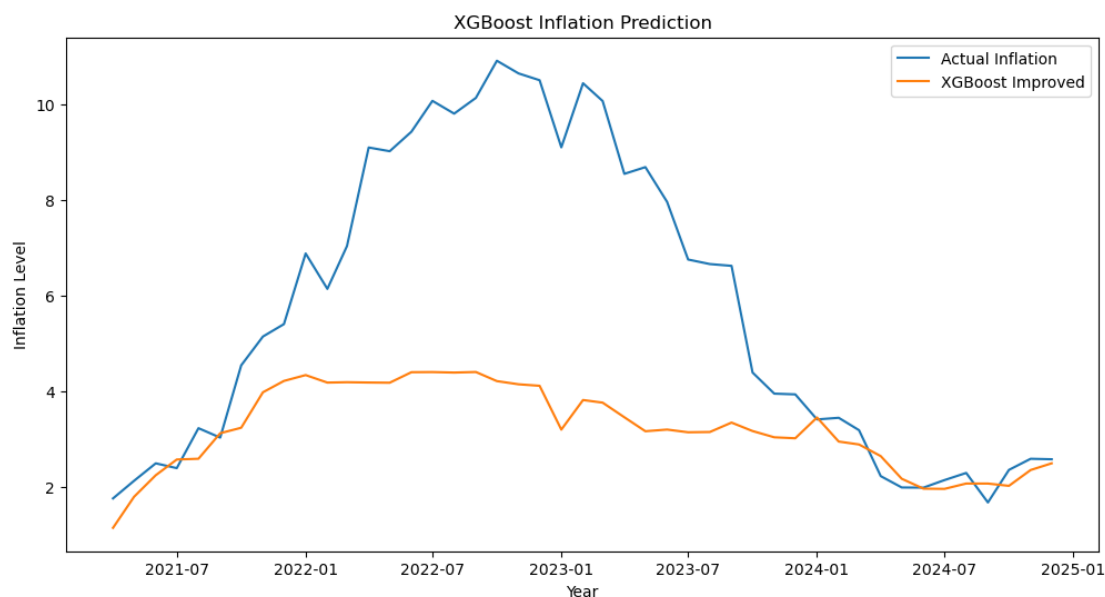


Figure 21: XGBoost Actual vs Predicted Inflation

The forecasting results show that the prediction performance of XGBoost model is good in several economic periods. The model was able to capture the overall direction and major fluctuations of the inflation behaviour better than the ARIMA baseline model.

When the economy was more stable, the forecasts based on XGBoost followed the actual inflation process more closely and were more sensitive to shifts in the inflation process. It also demonstrated a stronger ability to handle non-linear macroeconomic relationships than traditional statistical forecasting approaches.

However, as with the Random Forest model, forecasting was constrained during periods of extreme inflation volatility. XGBoost was able to capture parts of the inflation surge observed during 2022–2023, but some underestimation of peak inflation values and fast structural changes was still visible.

The forecasting performance of the XGBoost model was evaluated using MAE and RMSE metrics.

Table 7 shows the forecasting performance achieved by the XGBoost model.

Metric	Score
RMSE	2.77
MAE	2.55

Table 7: XGBoost Forecasting Performance Summary

The XGBoost model achieved forecasting performance superior to the ARIMA baseline but slightly weaker than the Random Forest model implemented in this study.

The results indicate that boosting-based machine learning methods can effectively model complex inflation behaviour in macroeconomic datasets. But the model also

showed more sensitivity to structural economic disruptions and validation windows than Random Forest.

This behaviour could be related to the highly optimised sequential learning structure of XGBoost that can become sensitive to abrupt regime shifts and sudden deviations of economic behaviour from the past.

From a broader forecasting perspective, the XGBoost implementation further confirms the usefulness of machine learning techniques in inflation forecasting applications, particularly in modelling non-linear relationships and interacting macroeconomic variables.

The XGBoost model showed good forecasting ability and served relevant comparative insight for the performance of advanced ensemble learning techniques for macroeconomic prediction tasks.

5.5 Comparative Performance Analysis

This section presents a comparative assessment of the forecasting performance of the ARIMA, Random Forest and XGBoost models implemented in the present study. The purpose of the comparison is to find the forecasting approach that exhibited the best predictive power and robustness under different macroeconomic conditions.

The forecasting models were evaluated using:

- Mean Absolute Error (MAE),
- Root Mean Square Error (RMSE),
- and comparative behavioural analysis of actual versus predicted inflation trends.

Table 8 shows the comparative forecasting performance achieved by the three implemented models.

Model	RMSE	Insight
ARIMA (Baseline)	3.88	Underfits
Random Forest	2.64	Best
XGBoost	2.77	Slightly Unstable

Table 8: Comparative Forecasting Performance of Models

The comparison results show that the machine learning models performed significantly better than the baseline ARIMA model for all the implemented forecasting periods.

The Random Forest model is the best forecast model of all models, with the smallest values of the forecasting error and more stability through volatile economic periods. The model resulted in an RMSE of around 2.64, which is a huge improvement compared to the ARIMA baseline RMSE of around 3.88.

The XGBoost model also had good predictive ability and was better than the ARIMA model. However, its forecasting performance was still somewhat inferior to Random Forest, especially during periods associated with structural economic disruption and extreme inflation volatility.

The comparative analysis additionally revealed clear differences in the model behaviour during the inflation peak of 2022-2023. All models struggled to forecast during this period, but the machine learning approaches showed better sensitivity to shifting inflation dynamics than ARIMA.

ARIMA model produced relatively stable, but very conservative forecasts and could not sufficiently capture fast inflation escalation during crisis periods. Random Forest and XGBoost, on the other hand, were more flexible as they incorporated multiple macroeconomic indicators and engineered features simultaneously.

The improved performance of the machine learning models was likely due to several factors:

- incorporation of macroeconomic predictor variables,
- feature engineering techniques,
- ability to model non-linear relationships,
- and improved flexibility under changing economic conditions.

Figure 22 presents a comparative visualisation of forecasting performance across the implemented models.

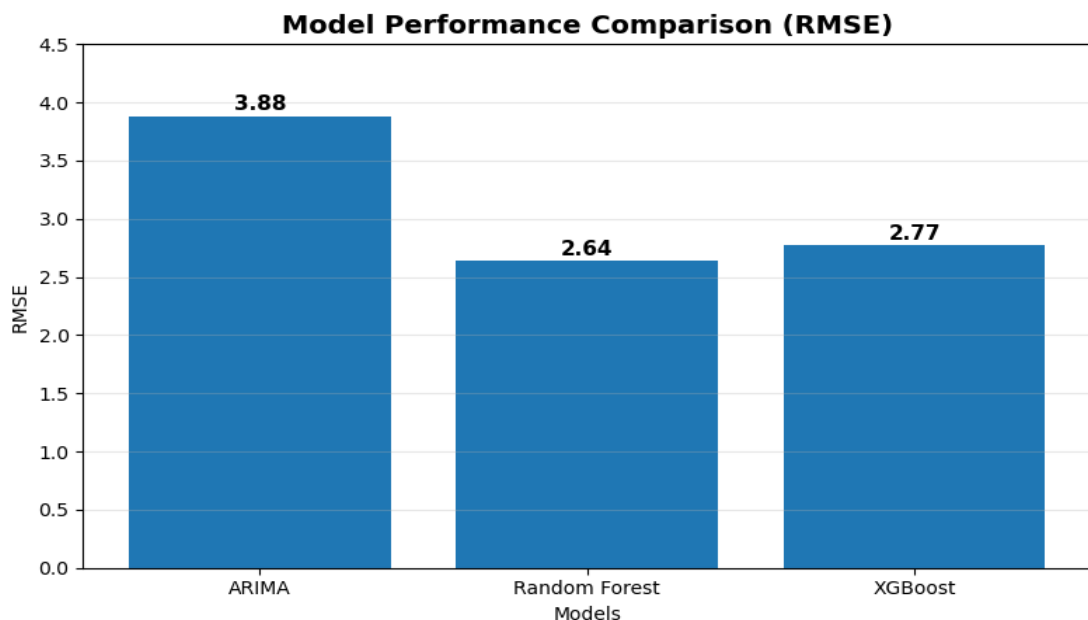


Figure 22: Comparative Forecasting Performance Across Models

From a wider forecasting perspective, the comparative analysis points to machine learning approaches being more appropriate for modelling inflation behaviour in

complex and volatile macroeconomic environments than traditional statistical forecasting models.

The results also underscore that even the most advanced machine learning models are still challenged to perform well during episodes of severe structural economic disruption and extreme inflation spikes, though the forecasting performance is improved.

In general, the comparison test proved that Random Forest was the most robust forecasting method throughout this study and that it was the most efficient forecasting method among the models applied.

5.6 Validation and Robustness Analysis

To further assess forecasting reliability, additional validation and robustness analysis was carried out across multiple economic periods. Since inflation behaviour can vary considerably under different macroeconomic conditions, relying on a single validation split may not provide a complete understanding of a model's ability to generalise over time.

For this reason, the study implemented:

- multiple validation windows,
- and walk-forward validation techniques

to examine forecasting consistency under both stable economic conditions and periods of high volatility.

The robustness analysis aimed to evaluate:

- how consistently the models performed across time,
- how sensitive they were to structural economic shifts,
- and whether forecasting accuracy remained stable under different validation settings.

Table 9 summarises the validation windows implemented within this study.

Validation Window	Training Period	Testing Period	Purpose
Validation 1	2000–2018	2019–2022	Evaluate forecasting behaviour during transition into COVID-19 disruption and early inflation surge
Validation 2	2000–2022	2023–2024	Assess forecasting stability under

			recent post-inflation economic conditions
Validation 3	2000–2019	2020–2024	Evaluate robustness during highly volatile crisis-related economic periods

Table 9: Validation Window Configuration

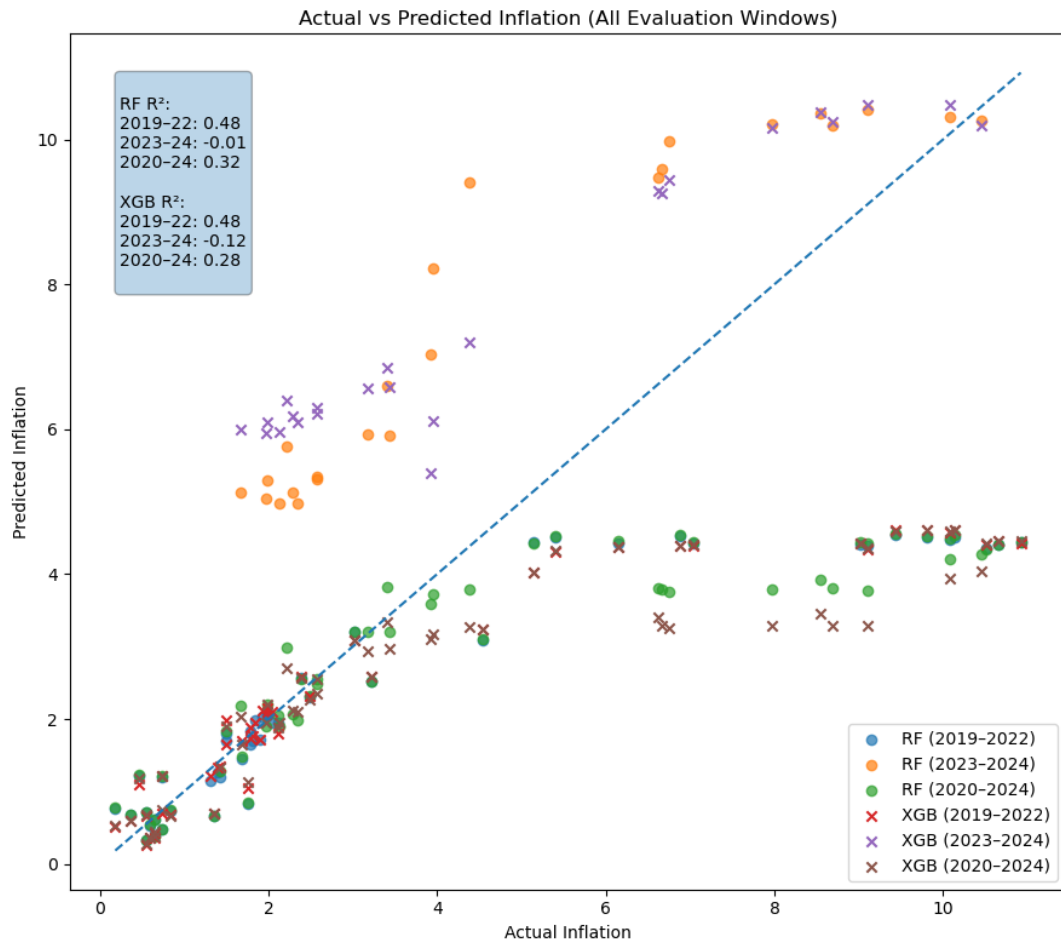


Figure 23: Actual vs Predicted Inflation Across Validation Windows

The validation results show that forecasting performance changed in different economic environments. During stable economic periods, both Random Forest and XGBoost produced forecasts that were closer to actual inflation values. However, the deviations in forecasts increased significantly during times of structural economic disruption and high inflation volatility.

The results also reveal that the Random Forest model showed greater consistency in forecasting over several validation windows. Specifically, it achieved better R^2 performance during the 2019–2022 and 2020–2024 evaluation periods compared to XGBoost.

Both machine learning models faced challenges in forecasting during periods of severe inflation increases, especially during the inflation surge after the pandemic. This behaviour underscores the forecasting difficulties caused by structural breaks and rapidly changing macroeconomic conditions.

Despite these challenges, the machine learning methods were much more flexible than the ARIMA baseline model. They showed a better ability to model complex macroeconomic relationships across different economic conditions. In addition to fixed validation windows, walk-forward validation was used with TimeSeriesSplit cross-validation techniques to further test forecasting strength over sequential forecasting periods.

Walk-forward validation simulates real-world forecasting behaviour by:

- training the model on an initial historical period,
- generating forecasts for unseen future observations,
- expanding the training dataset,
- and repeating the forecasting process across the dataset.

This validation method gave extra insights into forecasting stability and helped lower the risk of overestimating model performance during isolated validation conditions.

Thus, validation and robustness analysis showed that machine learning forecasting methods provided better consistency and greater adaptability in volatile macroeconomic environments compared to traditional statistical forecasting methods.

5.7 Structural Break and Crisis Forecasting Discussion

One of the key findings of this study was how structural economic disruptions affected forecasting performance. Structural breaks happen when historical economic relationships change significantly over time due to major external shocks, policy changes, or unusual macroeconomic conditions. These disruptions create instability in time series data and complicate forecasting.

The results from this study show that forecasting performance worsened during times associated with major economic crises, particularly:

- the COVID-19 pandemic,
- and the post-pandemic inflation surge seen between 2022 and 2023.

During these times, inflation behaviour deviated significantly from historical trends. Inflation volatility increased quickly, and macroeconomic relationships became less stable due to shifting economic conditions, supply chain issues, job market instability, and geopolitical tensions.

The forecasting models used in this study showed different levels of sensitivity to these structural disruptions. The ARIMA model faced the most challenges because it relied on stable historical patterns and linear time series assumptions. The results

revealed that ARIMA produced flat predictions and significantly underestimated the severity of the inflation surge.

While the machine learning models were more adaptable, they still had limitations during periods of extreme inflation. Both Random Forest and XGBoost responded better to changing inflation dynamics than ARIMA, but they too underestimated peak inflation levels.

This pattern suggests that even advanced machine learning models can struggle when:

- economic conditions change quickly,
- historical relationships become unstable,
- and highly unusual events are not well represented in the training data.

The results also highlight the need for robust validation strategies in macroeconomic forecasting research. Testing forecasting models across various economic environments offered a clearer picture of model behaviour compared to only using one validation period.

From a wider economic view, the findings suggest that inflation forecasting gets harder during crisis-related situations because:

- macroeconomic interactions become more volatile,
- policy responses change rapidly,
- and economic behaviour shifts faster than historical patterns can account for.

The structural break analysis further emphasizes the importance of:

- feature engineering,
- machine learning adaptability,
- and scenario-based forecasting methods in modern economic forecasting systems.

Despite the forecasting challenges posed by structural disruptions, the machine learning models used in this study showed better forecasting ability and reliability compared to traditional statistical methods.

The findings indicate that while forecasting models under changing macroeconomic conditions, their accuracy remains affected by major economic disruptions and times of extreme inflation volatility.

5.8 Chapter Summary

This chapter described the implementation, evaluation, and validation of forecasting models developed for inflation prediction in this study. Three forecasting approaches were implemented and compared: ARIMA, Random Forest, and XGBoost.

The chapter showed the ARIMA model was useful at capturing the general inflation behaviour in comparatively stable economic times and had clear limitations in times

of structural economic change. The machine learning models, on the other hand, showed a significantly better forecasting performance because they were able to consider several macroeconomic indicators and model the non-linear relationships in the data.

Random Forest (RF) model had the best performance in the forecasting of all the models while the comparatively higher robustness was shown by the same model in the various validation environments. The forecasting performance of XGBoost was also very good, but slightly more sensitive to structural economic changes than Random Forest.

The chapter also provided comparative forecasting assessment with various validation windows and walk-forward forecasting methods. The findings showed that accuracy of forecasting was found to be different under different conditions of macroeconomic variables and especially under conditions linked to the COVID-19 pandemic and the inflation surge in 2022-2023.

Moreover, the key forecasting issues posed by structural breaks, economic disruptions during crises and the very fast evolution of inflation were identified. While machine learning models performed better in predicting inflation compared to traditional statistical methods, predicting extreme inflation events was still challenging for all the implemented models.

The results offered in this chapter provide evidence of the effectiveness of machine learning forecasting methods when applied to macroeconomic forecasting applications and highlight the need for robust validation and forecasting methods in the context of a crisis.

The next chapter describes the creation of the interactive dashboard and simulation framework for converting outputs of forecasting into more meaningful economic risk insights and decision-support applications.

6. DASHBOARD AND SIMULATION TOOL

6.1 Introduction

This chapter provides a description of the development of the interactive dashboard and simulation framework that will be used to assist in the monitoring of inflation, the interpretation of inflation crisis and scenario-based economic analysis in this study. To make the forecasting results of the machine learning models interpretable, the dashboard system was designed to convert the forecasts into visual displays that can be understood by managers and decision makers.

Numerical predictions are sometimes produced by traditional forecasting models that might be hard to use for non-technical applications. In this study, the focus is more than just implementation; predictive outputs are made available via interactive business intelligence and simulation environments.

This study has resulted in the creation of two dashboard platforms:

- Power BI
- Streamlit

Both platforms allowed for comparison of traditional business intelligence reporting tools with more flexible interactive applications written in Python. Power BI offered comprehensive visualisation and reporting functions whereas Streamlit allowed for greater flexibility in scenario simulation, interaction with dynamic forecasting and user analysis.

The Dashboard framework was created to enable:

- inflation trend monitoring,
- forecasting visualisation,
- crisis risk classification,
- model performance interpretation,

They use the scenario simulation approach both at the micro and macro level.

The forecasting results from the Random Forest and XGBoost models were added to the dashboard systems along with:

- confidence interval estimates,
- risk classification measures,
- feature importance outputs,
- and macroeconomic indicators.

The key focus was to make it easier to interpret and use for non-technical users. As such, colour-coded risk indicators, KPI cards, interactive filtering capabilities and scenario-based forecasting capabilities were included in the dashboard environment.

The chapter also explores:

- dashboard architecture,
- simulation workflow,
- user interaction mechanisms,
- and applications of the forecasting system in business intelligence and economic decision-support settings.

The implementation and functionality of the Power BI dashboard and Streamlit simulation framework that were developed in this study are presented in the following sections.

6.2 Power BI Dashboard Implementation

Power BI was initially chosen as the prime dashboard platform, as it has solid business intelligence features, allows for interactive visualisation solutions, and can be used to provide forecasting knowledge in an executive-style reporting structure. The dashboard aimed to offer an accessible interface to present users with a view of inflation behaviour, to analyse forecasting results, and to interactively interpret the economic risk levels.

The Power BI dashboard related to outputs from the forecasting pipeline, including:

- actual inflation values,
- predicted inflation values,
- confidence interval estimates,
- crisis risk classifications,
- and macroeconomic indicators.

The datasets exported from the Python forecasting environment were used to develop the dashboard architecture.

Figure 24 presents the architecture of the Power BI dashboard system.

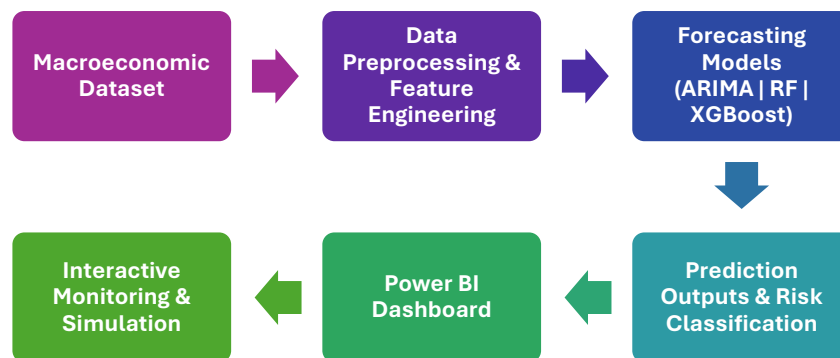


Figure 24: Power BI Dashboard Architecture

Several interactive visual elements were added to the dashboard to enhance interpretability and aid economic monitoring including KPI cards, inflation forecasting visualisations, risk-level indicators, confidence interval charts and date and interactive filters.

The KPI area presented some key forecasting data such as:

- latest actual inflation,
- latest predicted inflation,
- prediction error metrics,
- and present economic risk classification.

The interactive line charts provided a visual comparison of actual inflation behaviour to forecasting behaviour over time were used. Risk classifications were represented using colour-coded indicators to improve interpretability and provide rapid identification of changing economic conditions.

Forecasted Inflation Level	Risk Classification	Dashboard Indicator Colour
Inflation < 3%	Low Risk	Green
3% ≤ Inflation < 5%	Moderate Risk	Yellow
5% ≤ Inflation < 8%	High Risk	Orange
Inflation ≥ 8%	Crisis Level	Red

Table 10: Inflation Risk Classification Framework

Instead of manually guessing what the new inflation forecasts mean, the dashboard automatically transforms forecasts into insights by sorting them into clear risk tiers. The color-coding makes a huge difference, letting the user catch shifting trends before they become a problem.

The main Power BI dashboard interface created in this study is shown in Figure 25 and 26.

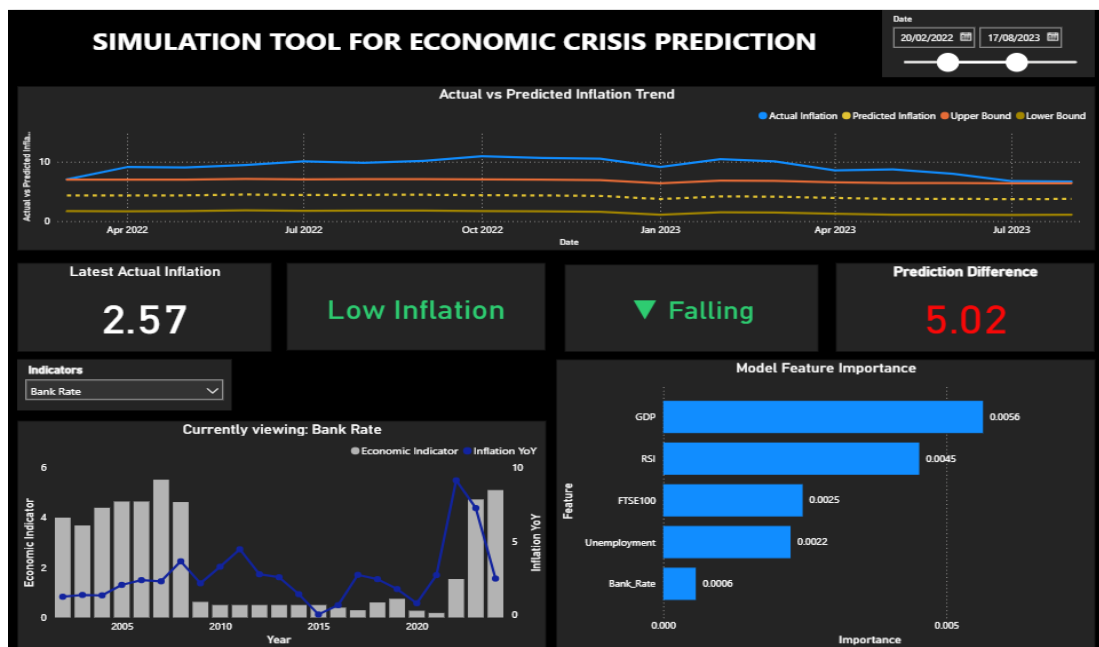


Figure 25: Power BI dashboard interface



Figure 26: Power BI dashboard interface (Crisis Simulator)

Users were able to dynamically explore inflation behaviour across different economic periods using interactive filtering and date-slicer functionality. Interactive filtering and date-slicer functionality enabled users to explore inflation behaviour dynamically across different economic periods. This allowed for more flexible analysis of forecasting performance in:

- stable inflation environments,
- COVID-19 disruption periods,
- post-pandemic inflation surge.

The Power BI dashboard was a success and demonstrated the ability to effectively convert the forecasting results into a user-friendly business intelligence environment that can assist in the monitoring of inflation and interpretation of economic risks.

6.3 Streamlit Simulation Framework

In addition to the Power BI dashboard, a Streamlit-based simulation framework was created to expand the interactivity and the ability to simulate scenarios in the forecast. Streamlit is chosen for its ability to seamlessly embed Python-based forecasting logic, interactive controls, and dynamic visualisation in a single analytical framework.

The Streamlit app was created for the following purposes:

- inflation forecasting,
- crisis risk monitoring,
- macroeconomic scenario simulation,
- and interactive economic analysis.

Figure 27 shows the architecture of the Streamlit simulation framework built in the scope of this study.

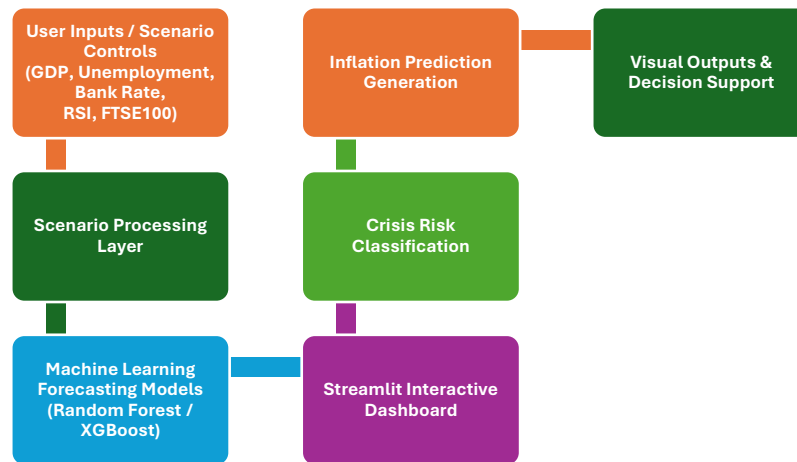


Figure 27: Streamlit Simulation Framework Architecture

The framework incorporated the forecasting results of the machine learning models and the user-controlled simulation capability. Users could vary the macroeconomic parameters and see the evolution of the behaviour of inflation and the level of crisis risk in a dynamic way in the future as a result of the evolution of the economic parameters.

The simulation environment included modifiable macro-economic parameters such as:

- GDP growth,
- unemployment rate,
- Bank Rate,
- retail sales behaviour,
- and the financial market indicators.

The system generated the following:

- projected inflation forecasts,
- inflation confidence ranges,
- and Dynamic crisis risk classifications.

The main Streamlit simulation dashboard interface built as part of this study is shown in figure 28 and 29.

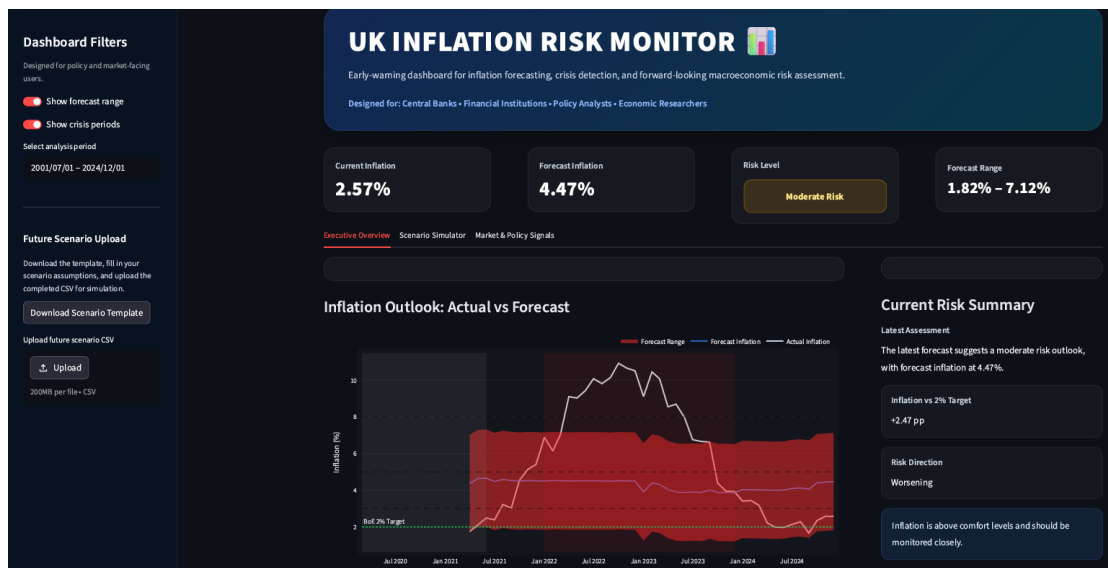


Figure 28: Main Interface of Streamlit Inflation Risk Simulation Dashboard

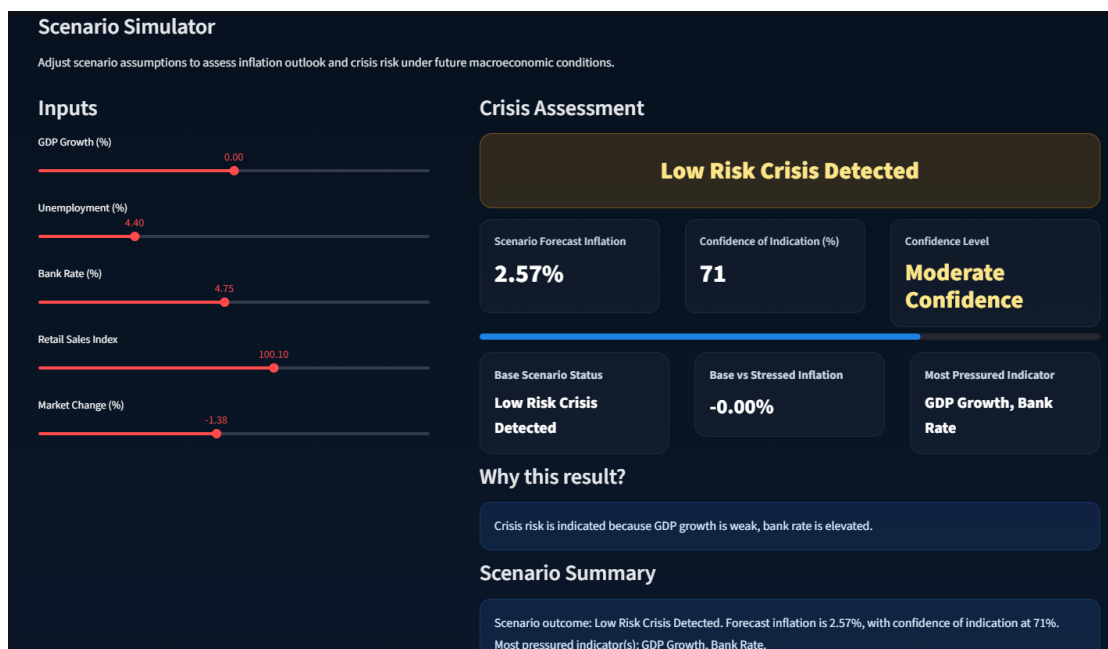


Figure 29: Dashboard Scenario/Risk Simulator

To ensure a user-friendly and easily understood forecasting process, several interactive features were developed and integrated into the Streamlit environment. These included:

- slider-based scenario controls,
- dynamic forecasting updates,
- interactive charts,
- confidence indicators,
- and risk monitoring indicators.

The framework also allowed users to provide future scenarios for macroeconomic conditions and study the impact of varying economic conditions on inflation outlook and crisis risk. This feature enabled the users to try out a scenario that might have occurred given different economic conditions and see forecasting behaviour, given different macroeconomic assumptions.

Compared to traditional dashboard environments, Streamlit was much more flexible in embedding the forecasting logic and interactive simulation directly into a Python workflow. This enhanced the responsiveness and adaptability of the forecasting environment and facilitated more in-depth analysis by users.

In summary, the Streamlit simulation framework enhanced the impact of this study by offering a flexible and interactive decision-support environment for inflation monitoring and macroeconomic scenario evaluation, well beyond the static forecasting analysis.

6.4 Practical Applications of the System

The forecasting/simulation system created in this study offers some practical applications in an environment of economic monitoring and business intelligence. The framework combines forecasting tools with interactive dashboard systems to make inflation and economic risk analysis more accessible, more interpretable.

The system can accommodate inflation monitoring, macroeconomic trend analysis, crisis risk evaluation and scenario-based forecasting.

The dashboard allows the user to explore the impact of changes in economic activity on the future inflation trend and risk profile. This capability could be helpful for:

- policymakers,
- financial analysts,
- researchers
- and business intelligence professionals.

The model framework can also be used to explore and study the educational value, as users can try and adjust macroeconomic scenarios and see how they affect the inflation forecasts.

Moreover, the incorporation of forecasts in visual dashboards enhances the interpretability of the model results, by translating model predictions into user-friendly economic insights.

The system provides a good example of how machine learning forecasting tools and business intelligence tools can be integrated to enhance the interactive economic analysis and decision support applications.

6.5 Dashboard and System Limitations

The forecasting and visualisation abilities achieved by the developed framework were found to be good, but there are still some limitations that were found during implementation.

A restriction was the reliance on external forecasts and pre-processing. The forecasts models and dataset were run in the Python environment prior to their integration into the dashboard systems.

Further, real-time forecasting was restricted as there were no fully automated, live economic data feeds. Consequently, there was a need for manual refresh or execution procedures every so often for the updated macroeconomic datasets and macroeconomic forecasting outputs.

In contrast to the Streamlit framework, Power BI environment also had limitations in user-driven simulation and large-scale forecasting interaction.

In addition, forecasters' accuracy was also found to be vulnerable to structural economic shocks and high inflation volatility especially when there was a crisis such as the inflation episode of 2022–2023.

Another drawback was the use of historical macroeconomic relationships. Historic data alone may not be enough to fully explain how unexpected geopolitical events, policy changes and external economic shocks may affect inflation behaviour.

The dashboard helps make forecasting outputs easier to understand, but it is not a substitute for economic expertise or policy analysis. Forecasts should be viewed in the context of the broader economic evidence.

In conclusion, the developed framework showed promising forecasting and decision-support abilities, but the results indicate that the models must be continuously improved, real time data integration is needed, and adaptive forecasting methods are essential for future economic prediction systems.

6.6 Chapter Summary

In this chapter, the development and implementation of the dashboard and simulation framework used in this study for supporting interpretation of the forecasts of inflation and assessment of economic risks were presented.

The chapter covered embedding forecasting outputs in the Power BI and Streamlit dashboard. The Power BI dashboard offered the facility to visualize and provide business intelligence on inflation monitoring, interpretation of the forecast, and assessment of the risk of crises. Furthermore, the Streamlit framework expanded the forecasting system to facilitate more flexible scenario simulation and to allow for analysis of the macroeconomic system based on user input.

The chapter also illustrated how the outputs of the forecasting process could be converted to meaningful dashboard information through:

- KPI monitoring,
- forecasting visualisation,
- risk classification,
- and a dynamic economic analysis that is interactive.

In addition, practical applications and the constraints of the developed system were discussed, and the potential advantages and problems with the forecasting deployment and economic simulation were summarized. In general, the dashboard and simulation framework built on this study were intended to go beyond the normal forecasting study by incorporating interactive business intelligence and decision support function or activities.

The final conclusions and the summary of the main findings of the study are presented in the following chapter, as well as the discussion of the research limitations and recommendations for further research.

7. CONCLUSION AND FUTURE WORK

7.1 Research Summary

This research paper proposed a framework of using machine learning and business intelligence techniques for inflation forecasting and economic crisis predicting. The study consolidated the macroeconomic forecasting, classification of crisis risk, validation analysis and construction of an interactive dashboard in one analytical system.

The data was taken from the UK, and the data used included CPI, GDP, unemployment, Bank Rate, FTSE100 and Retail Sales Index, which were all taken monthly from 2000 until 2024. After preprocessing and stationarity analysis, CPI was converted to target variable “Inflation_YoY” which was used for the entire forecasting process.

Three methods of forecasting were used and compared:

- ARIMA,
- Random Forest
- and XGBoost

Results showed that the machine learning models significantly performed better than the ARIMA baseline model especially during the transition of the macroeconomic environment. In terms of forecasting accuracy and relatively high forecasting robustness that was achieved in several validation environments, the Random Forest method was found to be the best.

Errors in forecasts were also attributed to structural economic disruptions, as observed in some scenarios, such as the COVID-19 shock and the inflationary shock of 2022-2023. All models used to predict extreme inflation peaks were not very successful in forecasting capacity, but machine learning models did improve their ability to predict capacity.

Further developments, expanded the forecasting interpretability and practical utility to:

- Power BI and Streamlit dashboard systems,
- mechanisms for the classification of crisis risk,
- And a scenario-based simulation function.

The forecasting outputs enabled the conversion to a business intelligence and decision support environment, where it can be used for economic monitoring and scenario analysis. The findings suggest that the potential of utilizing machine learning forecasting methods along with business intelligence tools for forecasting and analysing economic risks and inflation.

7.2 Achievement of Research Objectives

The development of an integrated inflation forecasting and economic crisis prediction framework was the main objective of this study and has been successfully accomplished.

The research successfully:

- built and developed a UK macroeconomic dataset for forecasting inflation,
- Applied and compared ARIMA, Random Forest and XGBoost models,
- Assessed forecasting strength in several times periods of the economy,
- Created interactive Power BI and Streamlit dashboards for risk monitoring and scenario simulation.

The findings suggest that the machine learning methods, specifically, the Random Forest model outperformed the ARIMA baseline model in forecasting performance and the dashboard framework enhanced the interpretability and usability of forecasting in practice.

This proved that the use of machine learning forecasting methods together with business intelligence tools were effective in predicting inflation and conducting economic risk analysis.

7.3 Key Findings

The results of this study suggest machine learning methods are more effective in forecasting inflation than traditional statistical methods. The best performance was achieved by Random Forest, and it was seen to be relatively more robust in different validation environments among implemented models.

The research also revealed that forecasting inflation is much more complex when it occurs during structural economic disruption and extreme volatility. The machine learning models achieved better results in capturing the inflation behaviour than the ARIMA but none of machine learning models was able to predict well during the inflation spike in 2022-2023.

The exploratory analysis also verified that the following were important:

- lag-based inflation behaviour,
- macroeconomic interactions,
- and engineered features

within inflation forecasting applications.

Furthermore, the forecasting results were mapped to Power BI and Streamlit dashboards which made the forecasts more interpretable and enabled interactive tools to observe the economy, as well as assist decision making.

The interesting result was that the Random Forest always performed better than the ARIMA and XGBoost for the various validation windows. This suggests that ensemble tree-based models may deliver more satisfactory robustness to the shifting macroeconomic environment and forecast accuracy. While Random Forest was like XGBoost, it appeared to be less affected by structural changes identified during the crisis periods.

In conclusion, the study highlighted the importance of integrating machine learning forecasting models with business intelligence systems in the context of predicting inflation and analysing economic risks.

7.4 Limitations of the Study

The forecasting framework was designed, and evaluated, however, there were several limitations identified as the following:

The models were based mainly on the relationships between variables in the past, and forecasting errors will occur if unforeseen external shocks or sudden structural economic changes occur. Events not well captured in the historical data, such as: Geopolitical instability, Energy price crises, and Rapid policy changes can affect inflation behaviour.

The machine learning models were found to be more effective in predicting the performance than the baseline ARIMA model, however extreme inflation peaks were difficult to predict for all the machine learning models used.

There was a limitation of the study because the data used were macroeconomic data and not very frequent. Future research should utilize other factors such as energy prices, sentiment indicators and real-time economic signals to improve forecasting accuracy.

Another restriction is the use of historical macro-economic relationships. Forecasting models have the assumption that the relationships found in the past are applicable to the future. These relationships have fluctuated rapidly during such economic events as the 2022–2023 inflation spike, which resulted in lower forecast accuracy. This accentuates the problem of modelling unusual economic events only based on previously available information.

Furthermore, the dashboard framework depended on the output of a third party forecaster and didn't contain any complete Realtime data integration by the dashboard.

The study was able to show the practical application of the use of machine learning forecasting models together with business intelligence and economic analysis based on simulation despite the limitations.

7.5 Future Work

There are several opportunities to extend and improve the forecasting framework developed in this study.

Future research could include:

- integration of real-time economic data,
- additional macroeconomic and geopolitical variables;
- and automatic forecast update systems.

Advanced deep learning methods such as:

- LSTM,
- transformer-based predictive models,
- and hybrid ensemble architectures.

can also be explored to enhance the forecasting performance in the presence of extremely volatile economic conditions.

Further enhancements to the simulation framework could include:

- deployment to the cloud,
- live API connectivity,
- and more sophisticated user-driven economic scenario analysis.

Furthermore, future research could expand the framework beyond UK inflation forecasting by incorporating multi-country economic datasets and comparative macroeconomic analysis.

Finally, further development of adaptive and Realtime forecasting systems can improve practical applicability of machine learning approaches in economic monitoring and crisis prediction environments.

7.6 Final Conclusion

The research designed a hybrid system for inflation forecasting and prediction of economic crisis based on machine learning and business intelligence techniques. The study demonstrated the ability to combine macroeconomic forecasting models with interactive dashboard systems to create interpretable economic monitoring and decision-support tools that can benefit from predictive analytics.

The outcomes showed that in different prediction validation scenarios machine learning techniques showed better performance compared to the baseline ARIMA model, especially with the Random Forest model. The study also highlighted the difficulty of forecasting the occurrence of structural economic shocks and high inflation volatility.

Besides forecasting implementation, Power BI and Streamlit dashboards enhanced the accessibility and practicality of the forecasting outputs by providing interactive visualization, categorization of crisis risks and simulation options based on scenarios.

The results showed the feasibility of combining the machine learning algorithm with the macroeconomic analysis and business intelligence frameworks for predicting inflation and assessing economic risk. The elaborated framework serves as a basis for the next steps of adaptive forecasting and economic monitoring systems.

LINKEDIN POSTS

Post 1

Pratiksha Jadhav • You
Data Analyst | Turning Data into Business Decisions | Power BI | Python | SQL | ...
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To what extent do macroeconomic indicators predict inflation and signal economic crisis risk?
That's the question I explored in my MSc dissertation at [University of Leicester](#) collaborated with [Uni worx](#)

But before building any model, I had to understand one thing:
How does inflation actually behave over time?

The chart below shows 20+ years of UK inflation (YoY).
At first glance, it looks relatively stable.

But when you layer in real-world context, the story changes completely:
COVID-19 shock (2020–2021) disrupted economic patterns
Inflation surge (2021–2023) led to one of the sharpest spikes in recent history

The same data — but a very different interpretation.

This was my first major realisation:
This isn't just a forecasting problem.
It's a pattern recognition problem shaped by economic shocks.

To explore this further, I worked with key macroeconomic indicators:
GDP, Unemployment, Bank Rate
FTSE100, Retail Sales
Inflation (CPI transformed to YoY)

Using Python (pandas, matplotlib, statsmodels), I built an EDA pipeline to:
Identify structural breaks in the data
Understand relationships between variables
Test assumptions like stationarity (ADF)
Explore lag effects across indicators

One challenge became clear very quickly:
The raw data wasn't ready for modelling.
- CPI showed strong trends → not stationary
- Relationships weren't obvious
- Key signals were hidden behind noise

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Figure 30: LinkedIn Post 1

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So I went back to fundamentals:
Transformed CPI → Inflation (YoY)
Focused on making the data statistically meaningful
Built a strong analytical foundation before modelling

Because I learned:
A good model starts with good understanding — not just good algorithms

This project stopped being "just a prediction task"
...and started becoming a data-driven approach to understanding economic risk

This is Phase 1 of my journey.

In the next post, I'll share:
How I built and improved forecasting models
What worked (and what didn't)
And how I tackled challenges like predicting inflation spikes

Curious to hear your thoughts —
Do you think economic crises can actually be predicted?

#DataAnalytics #DataScience #MachineLearning #Economics #TimeSeries #Python #PowerBI #OpenToWork

UK Inflation (CPI) and Crisis Periods (2020-2023)

Aakash Yadav and 14 others

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Figure 31: LinkedIn Post 1

Post 2

Pratiksha Jadhav • You
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3w • Edited •

The model seems to work... yet something isn't quite right...

This is what bothered me the most while constructing my inflation forecasting models.
Having finished with the data analysis, I started working on modelling – beginning with a basic model:

- ARIMA

This worked in capturing broad trends but fell short in its crucial task:
In situations of economic uncertainty

Therefore, I tried out machine learning algorithms:

- Random Forest
- XGBoost

From the calculations, it did look like the performance improved:

- ARIMA RMSE: 3.88
- Random Forest RMSE: 2.64
- XGBoost RMSE: 2.77

Clearly, Random Forest performed better here.

However, when I plotted the forecasts...

- The models still did not capture major inflation spikes in 2022–2023
- The predictions looked too smooth compared to reality
- Performance decreased in cases of changing economic circumstances

At this point, I realized something vital:
Prediction accuracy alone is not enough in unstable economies.
(This became my key insight.)

I returned, not to modify the architecture of the model, but to reconsider the input variables.

Here came the breakthrough of this phase:

- Lagged feature variables (1month, 3months, 6months, 12months)
- Inflation momentum and persistence captured
- Macro interactions engineered
- Standardised feature variables for stability

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Figure 32: LinkedIn Post 2

And most importantly:
I only concentrated on relevant macroeconomic drivers.

With the second iteration:

- Best-performing model was the Random Forest
- Performed much better in capturing all patterns
- Showed high prediction performance in stable periods

However, realized a big limitation that:
Models struggle when the economy shifts rapidly

This realization transformed my understanding of ML applications:
Not about achieving perfect forecasts,
...but about understanding when these can be trusted

This stage became crucial for me.
Moved from mere improving forecasts → to understanding Risk and Reliability.

In the next post – final one in this series,

- How I converted predictions into a crisis risk framework
- Built a Streamlit-based python simulation tool for real-world use
- And turned this into a decision-support system

Have you had similar experience, when making improvements did not help?

#MachineLearning #DataScience #RandomForest #TimeSeries #FeatureEngineering #Python #Analytics #MScdissertation

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Figure 33: LinkedIn Post 2

Post 3

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3d ·

Building forecasting models was only half the journey.

The bigger challenge was:

- How can economic predictions become something people can actually interact with and use?

That led to the final stage of my MSc dissertation:

- Developing an interactive UK Inflation Risk Monitor & Economic Crisis Simulation Tool using Streamlit.

Initially, I explored Power BI for dashboard development. But as the project evolved, I needed something more flexible for:

- Scenario simulations
- Dynamic forecasting
- User-driven inputs
- Future scenario uploads

So I transitioned to a Python-based Streamlit application — and that completely transformed the direction of the project.

The final system allows users to:

- Monitor inflation trends and forecast ranges
- Analyse crisis risk levels in real time
- Upload future macroeconomic scenarios
- Simulate how economic conditions may affect inflation outlook

Users can experiment with variables such as:

- GDP Growth
- Unemployment
- Interest Rates
- Retail Sales
- Market Conditions

...and instantly observe how crisis risk levels change under different economic assumptions.

To improve interpretability, I also designed a threshold-based risk framework:

- Low Risk
- Moderate Risk

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Figure 34: LinkedIn Post 3

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Figure 35: LinkedIn Post 3

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APPENDIX

Table A1. Data Sources

Variable	Source
CPI (Inflation)	ONS
GDP	ONS
Unemployment	ONS
Bank Rate	Bank of England
FTSE100	Yahoo Finance
Retail Sales Index	ONS

Table 11: A1. Data Sources

Table A2. Engineered Features

Feature	Description
Inflation_lag1	Previous month's inflation
Inflation_lag2	2-month lag
Inflation_lag3	3-month lag
Inflation_lag6	6-month lag
Inflation_lag12	12-month lag
GDP_growth	Monthly GDP growth
FTSE_returns	Monthly FTSE return
RSI_change	Monthly retail sales change
Rate_Unemployment	Interaction feature
Market_Sentiment	FTSE return $\times$ RSI change

Table 12: A2. Engineered Features

Table A3. Final Model Parameters

- Random Forest:

Parameter	Value
n_estimators	1200
max_depth	20
min_samples_split	3
min_samples_leaf	1

Table 13: A3. Final Model Parameters (RF)

- XGBoost:

Parameter	Value
n_estimators	2000
Learning_rate	0.01
max_depth	6
subsample	0.8
colsample_bytree	0.8

Table 14: A3. Final Model Parameters (XGB)

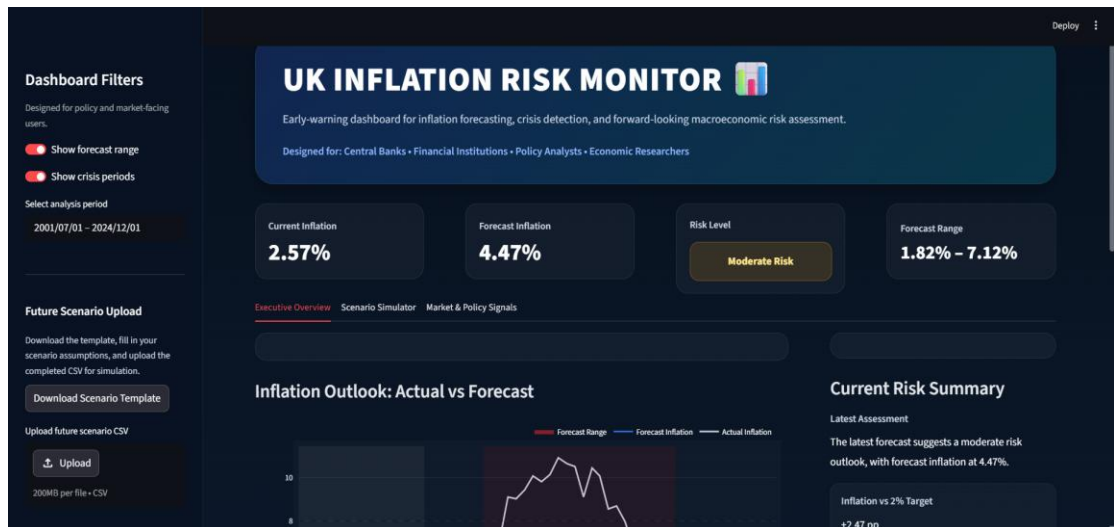


Figure 36: A1. Final Streamlit-based Economic Crisis Prediction Dashboard.